S^{2}Net: Spatial-Aligned and Semantic-Discriminative Network for Remote Sensing Object Detection

Jiayu Yao, He Chen, Yizhuang Xie, Ning Zhang, Mingxu Yang, and Liang Chen

***Abstract*—With the development of remote sensing technology in recent times, those geographically located objects such as vehicles, ships, airplanes, and oil tanks have become one of the most important information resources in many civilian applications. However, such remote sensing images (RSIs) are still struggling with the problems of huge scale variation and limited semantic details. To tackle these challenges, a spatial-aligned and semantic-discriminative network, S^{2}Net, is proposed. First, grouped spatial attention-based fusion (GSAF) is designed for balanced and spatial-aware multiscale feature fusion. Second, a loss function for box regression, edge-ratio IoU (ER-IoU) loss function, is presented to improve convergence speed and localization accuracy, which explicitly optimizes the edge-to-edge alignment between anchor and ground-truth boxes, followed by aspect ratio calibration. Finally, semantic-adaptive contrastive learning (SACL) is proposed, which brings similar instances closer together and pushes dissimilar instances farther apart with straightforward alignment of the content queries. Experiments on three public datasets NWPU VHR-10, DIOR, and RSOD demonstrate that S^{2}Net achieves superb accuracy benefits compared with recent advanced methods, attaining 96.3%, 81.4%, and 96.8%, respectively.**

***Index Terms*—Box regression, contrastive learning, detection transformer (DETR), multiscale feature fusion, remote sensing object detection (RSOD).**

I. Introduction

REMOTE sensing object detection (RSOD) aims to determine whether or not objects of interest exist in a given remote sensing image (RSI) and to return the category and position of each predicted object [1]. As the cornerstone of many tasks, RSOD has a wide range of applications, such as urban and civilian applications, environmental or ecological monitoring, and agricultural and forestry applications [2], [3], [4], [5]. Exhibiting significant diversity in scale, angle, and appearance, RSIs cover widespread regions of Earth's surface. Thus RSOD has received significant attention and become increasingly critical.

Received 31 July 2025; revised 12 September 2025; accepted 13 October 2025. Date of publication 16 October 2025; date of current version 31 October 2025. This work was supported under Grant 2022YFB3902304. *(Corresponding author: Ning Zhang.)*

Jiayu Yao, He Chen, Yizhuang Xie, Mingxu Yang, and Liang Chen are with the National Key Laboratory of Science and Technology on Space-Born Intelligent Information Processing, Beijing Institute of Technology, Beijing 10081, China (e-mail: yjy@bit.edu.cn).

Ning Zhang is with the National Key Laboratory of Science and Technology on Space-Born Intelligent Information Processing, Beijing Institute of Technology, Beijing 10081, China, and also with the Department of Computing, Hong Kong Polytechnic University, Hong Kong (e-mail: nzhang.rs@bit.edu.cn).

Digital Object Identifier 10.1109/TGRS.2025.3622254

To detect objects in RSIs, early algorithms primarily relied on handcrafted features and classical classifiers, such as methods based on support vector machine (SVM) [6] and AdaBoost [7]. However, their robustness was limited due to a strong dependency on manually designed features and the inability to generalize across diverse remote sensing scenarios. With the rise and development of deep learning, convolutional neural networks (CNNs) have significantly transformed object detection frameworks. Two-stage detectors, including SPP-net [8], Faster R-CNN [9], and Mask R-CNN [10], improved accuracy by introducing region proposal mechanisms, but remain inefficient due to the need to produce candidate regions and the subsequent feature resampling stages. Meanwhile, one-stage detectors such as You Only Look Once (YOLO) [11], [12], [13], [14] and single-shot multibox detector (SSD) [15], [16] simplified detection as a regression problem by directly predicting object classes and bounding boxes from dense feature maps. These CNN-based models have provided powerful feature extraction capabilities and end-to-end training schemes. Nevertheless, there are several inherent shortcomings of CNN-based detectors. The fixed receptive field and local connectivity limit their ability when faced with complex backgrounds and distant targets.

In recent years, Transformer-based models have gained increasing attention in object detection due to their ability to capture long-range dependencies and establish global semantic relationships across the input image, leveraging the self-attention mechanism. However, their application often comes with high computational cost and slower convergence, which limits their practicality in real-time or resource-constrained scenarios. Detection transformer (DETR) [17] inventively introduces a hybrid model that combines convolutional backbone with transformer-based architecture. This design enables end-to-end object detection by eliminating handcrafted components, thus simplifying the pipeline and reducing heuristic dependencies. Subsequent improvements, such as Deformable DETR [18], introduce deformable attention modules to accelerate convergence and enhance feature aggregation, while DN-DETR [19] adopts denoising training strategies to further improve training stability and detection performance. To enhance efficiency further, RT-DETR [20] has been proposed as a fast and accurate variant, achieving real-time inference while maintaining high detection accuracy. Compared to traditional two-stage or anchor-based detectors, RT-DETR streamlines the detection process and significantly reduces latency, making it suitable for real-time applications. Nonetheless,

1558-0644 © 2025 IEEE. All rights reserved, including rights for text and data mining, and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.
Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 11:14:37 UTC from IEEE Xplore. Restrictions apply.

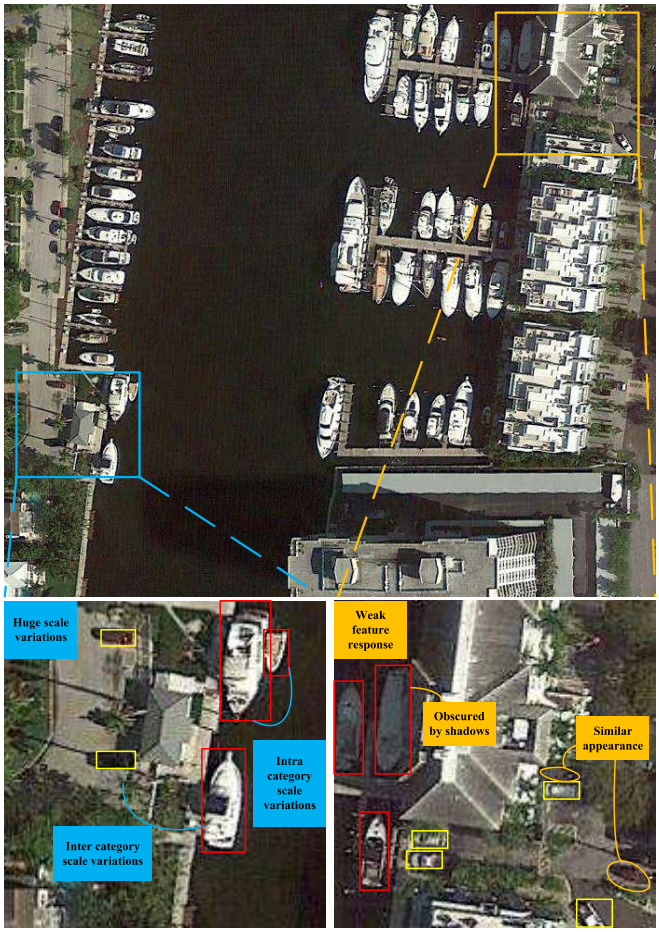


Fig. 1. Typical RSIs. (a) Huge scale variations exist within one scene. (b) Limited semantics and distortions lead to recognition interference.

the current body of work on DETR-based RSOD is limited and encounters several challenges.

1) In RSIs, the disparity in object scales is extreme. As illustrated in Fig. 1(a), objects as tiny as vehicles (just a few pixels) may co-occur with huge structures (airplanes and playgrounds) within one scene. This kind of scale variation severely degrades detector performance. Small targets occupy very limited spatial extent, yielding ambiguous feature maps with poorly defined edges and weak signals. Large targets, in contrast, span broad regions and demand large receptive fields to capture context. In consequence, when DETR-based models fuse multiscale features, they are prone to interfere with each other. Frequently, small objects are missed or poorly localized, and large objects may be split. Therefore, for the improvement of localization, models need to be equipped with the capabilities to handle multiscale objects.
2) Generally, some objects in RSIs have inherently low signal-to-noise ratios and limited semantic details, as shown in Fig. 1(b). These weak object features are further obscured by the complex imaging conditions: scenes often include cluttered backgrounds with geometric or radiometric distortions and cast shadows that mask or corrupt object cues. Besides, some objects may be overlapped with one another, which yields weak semantic activations and leads to undesirable performance. Considering that content queries are responsible for providing category semantic priors for DETR-based models, targeted improvements are advantageous for enhancing the precision of detection.

To address these issues mentioned above, a DETR-based detector S^2Net is proposed for multiscale RSOD. S^2Net mainly utilizes grouped spatial attention-based fusion (GSAF) to maintain balanced feature representation across all object scales, avoiding excessive downsampling or feature saturation for targets of any particular size. Meanwhile, edge-ratio IoU (ER-IoU) is designed and employed to accelerate bounding box regression and improve localization accuracy. Furthermore, semantic-adaptive contrastive learning (SACL) strategy is implemented for the refinement of category semantic priors. Extensive experiments are conducted on three public remote sensing datasets for the assessment of S^2Net: NWPU VHR-10 [21], DIOR [22], and RSOD [23]. The results reveal that S^2Net excels in detecting multiscale RSIs, with reduced parameter count and computational load. The main contributions of this study are summarized as follows.

1) A novel network for RSOD, S^2Net, is proposed, implementing spatial-aligned and semantic-discriminative modules or strategies. S^2Net first processes multiscale fusion by GSAF, combining features extracted from different receptive fields to gain both fine-grained details and broader contextual information. To optimize training dynamics, S^2Net employs ER-IoU that accelerates convergence and refines bounding box precision. Further enhancing its discriminative capability, S^2Net employs SACL, a mechanism that sharpens class-specific content queries' ability to isolate target features from cluttered backgrounds, even under challenging foreground–background overlap scenarios.
2) A GSAF module is proposed to tackle the challenge of fusing features from scale-diverse features. The main block grouped spatial attention (GSA) is designed to extract multiscale features with a dual-branch structure using different receptive fields. Pinwheel-shaped convolution (PConv) [24] modules are introduced to enhance local structural awareness. By employing GSAF, the detector performs well in processing scale-varying objects in RSIs.
3) An innovative bounding box regression loss, ER-IoU, is proposed. ER-IoU explicitly optimizes the edge-to-edge alignment between anchor and ground-truth boxes, followed by aspect ratio calibration. This loss function optimizes the model convergence and improves positioning precision.
4) An SACL strategy is proposed to provide semantic refinement for RSIs with limited semantic details, such as cluttered backgrounds. SACL combines interclass separation loss and intraclass compactness (ICC) loss with binary masks, enhancing the clarity of category semantic information in content queries to boost

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 11:14:37 UTC from IEEE Xplore. Restrictions apply.

detection performance. Without adding complexity to the model, SACL strengthens the robustness faced with corrupt object cues.

The rest of the article is organized as follows. Section II presents a review of the related works of RSOD. Section III outlines the proposed methods. Section IV details the experimental setup, followed by a comprehensive analysis of the results. Finally, Section V concludes the article.

II. Related Works

A. Object Detection In Remote Sensing

As a result of RSIs' geographical diversity, object scale, and morphology, RSOD is one of the most fundamental and challenging tasks and has received long-standing attention. Although there exists a plethora of methods of RSOD, there is still a lack of recent updates. Since CNN-based networks have found widespread applications in interpreting RSI, there is an increasing need for the balance of effectiveness and accuracy of RSOD. Existing work primarily focuses on network architecture design. Due to the excellent performance of YOLO series, many YOLO-based variants have been applied to RSOD. Zhao et al. [25] proposed SCENE-YOLO building upon YOLOv8, which introduces a scene information gathering and distributes network to inject high-level semantic information. A redesigned backbone incorporates the attention mechanism of omni-dimensional dynamic convolution to dynamically redistribute weights for target features. Subsequently, a scene label generation algorithm is proposed to supervise the model by generating scene-level labels. Finally, a scene-assisted detection head is introduced to enhance detection performance in complex backgrounds. Liu et al. [26] proposed MFAE-YOLO to tackle inadequate global feature extraction and unstable multiscale loss calculations. The approach introduces a global feature fusion processing module to enhance global feature extraction. The backbone network incorporates fusion of channel, pixel, and spatial attention modules to strengthen feature representation, while C2F-feature pool extraction units optimize pooling-based feature extraction. In addition, an accurate intersection over union loss function is proposed to refine bounding box regression. Besides, there are also numerous recent works that integrate multiple RSOD tasks or leverage multimodal images like infrared information to achieve better performance. Zhang et al. [27] proposed UniconDet to integrate multi-grained object detection requirements of HBB, OBB, and InSeg into one challenging task of arbitrary-shaped object contour detection. First, a Fourier contour parametric modeling is defined to project arbitrary-shaped object contours from the spatial domain into the frequency domain. Second, a multiview cross-attention feature extraction way is designed at each scale of the regression layer, and finally a center-contour enhancing regression layer is designed to generate regional guidance and cascade contour propagation, which can ensure a more accurate center point prediction and Fourier contour parameter regression. Liu et al. [28] proposed DPAL for multimodal RSOD. Particularly, a cross-modal alignment module is designed, which utilizes the multiscale dilation strategy and a differentiable alignment function with channelwise modulation for cross-modal feature integration. Additionally, a task-head alignment module is proposed. Wu et al. [29] proposed a transfer learning-based, large-scale, robust hyperspectral target detector (TLH2TD). First, the coupled target–background sample expansion module is designed to expand the labeled hyperspectral target and background samples with sufficient high-resolution, labeled RGB images, and a few labeled hyperspectral samples. Second, the hard positive and negative example mining module trains the hard positive and negative samples to enhance the discriminative ability of the network. Third, the spatial–spectral weighted subspace (SSWS) module is designed to fuse the spatial features extracted from the target detection network and the spectral features based on the Mahalanobis distance.

In the evolution of DETR-based models, a real-time and end-to-end object detector, RT-DETR [20], is proposed. An efficient hybrid encoder is designed to expeditiously process multiscale features, and an uncertainty-minimal query selection is proposed to provide high-quality initial queries to the decoder, juggling speed and accuracy. The work of RT-DETR has been applied to RSOD tasks. Wang et al. [30] proposed a real-time detector for ground-moving objects in optical RSIs, GMODet, which includes three novel modules. First, a region awareness module is designed to bolster the representation of global information from distant ground-moving objects and local information from adjacent pixels, as well as to implement diversified high-level semantic feature awareness strategies. Furthermore, the cross-scale context-aware feature aggregator is devised to aggregate valuable information contained in adjacent feature maps across different scales. Finally, the semantic–spatial progressive interaction module establishes a bridge between semantic and spatial information, enhancing guided learning for different categories of objects through the integration of coarse semantic information and shallow spatial features. Zhong et al. [31] proposed PSWP-DETR, which leverages adaptive deformation learning and multiscale integration for enhanced object detection in remote sensing. First, PradatorConv is proposed to address the significant shape changes of objects because it adaptively learns the horizontal and vertical deformations to perceive the complex geometric features of RSIs. Second, scalewise differential modules (SDMs) are proposed, which comprise multiscale convolution and edge captor convolution. SDM integrates features across various scales and captures edge characteristics and local textures. Finally, whale particle optimization algorithm is proposed for learning rate optimization.

However, the challenges of huge scale variations and weak feature responses posed by limited semantics and complicated backgrounds make it challenging for current RSOD methods to meet the demands in terms of low latency, high accuracy, and small model size. To further overcome the challenges of huge scale variations and weak feature responses caused by limited semantics and complicated backgrounds, S²Net is proposed, based on RT-DETR. S²Net proposes a comprehensive framework for RSOD by integrating three key technical innovations: the GSAF module to hierarchically aggregate multiscale spatial features, the ER-IoU loss function to

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 11:14:37 UTC from IEEE Xplore. Restrictions apply.

simultaneously optimize convergence efficiency and localization precision, and the SACL strategy to adaptively refine semantic priors. This unified architecture ultimately advances RSOD performance in both accuracy and robustness.

B. Multiscale Feature Fusion

Multiscale feature fusion is essential in RSOD due to the extreme variations in object size and background. Most CNN-based detectors employ a feature pyramid network (FPN) [32] or similar neck to combine high-level semantic features with lower level details. In practice, advanced variants like PANet [33], NAS-FPN [34], and BiFPN [35] add bottom-up paths or learned topology on top of FPN. Recent works introduce specialized fusion modules. For instance, Zhang et al. [36] proposed feature fusion module (FFM), an innovative lightweight and plug-and-play module to improve feature fusion strategy. FFM adopts a neck structure based on BiFPN, and reweight different feature maps by channel information without increasing computational complexity. The structure of FFM and its channel reweighting strategy optimize the fusion process of multiscale semantic information for small objects, which provides more effective feature maps for subsequent global context modeling. A multiscale feature pyramid can also be used in multimodal feature fusion. Wu et al. [37] proposed a multimodal adaptive FFM. High-weight features are adaptively fused, and low-weight redundant features are compressed to achieve adaptive fusion using a multiscale feature pyramid under different illumination conditions. Besides, some focus on global context and attention across scales to optimize fusion. Gao et al. [38] proposed GLSANet, which augments FPNs, and introduces a Local Attention Pyramid to amplify small-object features while suppressing the background and noise in the shallower feature maps. Multiscale structures have also been incorporated into transformer-based and hybrid models. Liu et al. [39] proposed an easy plug-in design, DeNoising FPN (DN-FPN), to suppress noise in each level's features in the top–down path of FPN, which copes with the problem that noisy features may be produced during the fusion process of the primitive FPN structure since there is no regularization between the features of different scales. For the CNN-based cross-scale feature fusion (CCFF) of RT-DETR [20], it is a combination of FPN and PANet, which utilizes fusion blocks consisting of convolutional layers to fuse two adjacent scale features into a new feature.

These architectures greatly improve fusion across scales, but they still struggle with the very large-scale disparities typical in RSIs. To address this, the GSAF module is proposed, which comprises the innovative GSA block and PConvs. By leveraging the multiscale feature extraction capability of the dual-branch GSA architecture and the enhanced local structural awareness provided by PConv, the GSAF framework demonstrates outstanding performance in processing complex multiscale scenes.

C. Loss Functions for Bounding Box Regression

For bounding box regression, the IoU loss [40] is well known for its achievement on face detection results, showing robustness to objects of varied shapes and scales. However, IoU has a plateau making it infeasible to optimize in the case of nonoverlapping bounding boxes. Consequently, Rezatofighi et al. [41] presented the GIoU loss, which calculates a ratio focusing on the empty volume between two boxes.

Moreover, faced with the problems of slow convergence and inaccurate regression, Zheng et al. [42] proposed the DIoU and CIoU losses, incorporating the normalized distance between the predicted box and the target box. To further improve both the speed of training and the accuracy of the inference, Gevorgyan [43] suggested the SIoU loss, where penalty metrics are redefined considering the angle of the vector between the desired regression. In current times, more methods to optimize the loss functions have poured out as well. Additionally, Zhang et al. [44] proposed the Diag-IoU loss to tackle the box inclusion dilemma of existing IoU-based losses in object detection. By exploiting the box diagonal information, the relationship between the predicted and target boxes can be well evaluated even in the case of box inclusions. Additionally, to address the issue that adjacent positive boxes often exhibit high levels of overlap in densely packed scenes full of crowded and small-sized objects, Wang et al. [45] designed the DeIoU loss, consisting of a dense box regression loss and a one-to-many label matching strategy guided by DeIoU. Notably, to avoid inappropriate expanding of anchor boxes, Liu et al. [46] proposed the PIoU loss, which combines a target size adaptive penalty factor and a gradient-adjusting function based on anchor box quality. It guides anchor boxes to regress along efficient paths, resulting in fast convergence.

These refinements for the loss function improve the performance of box regression, whereas most still struggle with the issues of anchor box enlargement. To manage this problem and to further refine the aspect ratio of bounding boxes, the ER-IoU loss function is proposed. By enforcing edge-to-edge alignment along with aspect ratio calibration, ER-IoU promotes the convergence behavior and regression precision.

D. Contrastive Learning

Contrastive learning [47] is a method by training a model to differentiate between similar and dissimilar samples, which has been shown to be effective and has gained significant attention in various computer vision and natural language processing tasks. It mainly can be categorized into self-supervised contrastive learning and supervised contrastive learning. Self-supervised contrastive learning includes classic methods like SimCLR [48], MoCo [49], PCL [50], and BYOL [51]. Supervised contrastive learning includes methods like Triple loss [52], SupCon loss [53], and EBM [54]. Recently, there have been many explorations on the selection of positive and negative samples of contrastive learning. For example, Jang and Wang [55] proposed a debiased contrastive learning method that addresses two types of biases: hard versus easy negatives and true versus false negatives (FNs). Furthermore, triple loss is introduced to amplify the bias in the representation space, which serves as an effective surrogate for learning the relative difficulty of samples in self-supervised contrastive learning. Tian et al. [56] proposed a sample weighting and screening

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 11:14:37 UTC from IEEE Xplore. Restrictions apply.

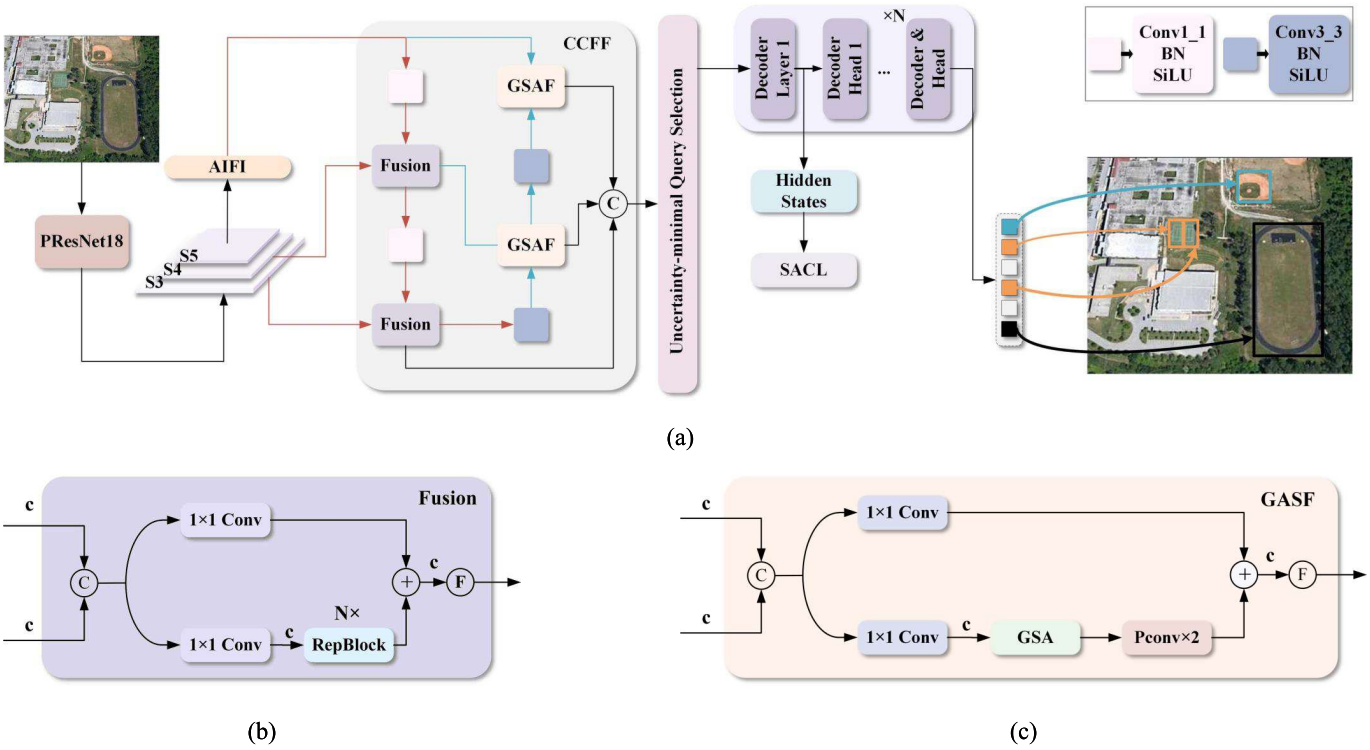


Fig. 2. Overview of S²Net. (a) Model uses backbone to extract features and keep the last three stages. The AIFI processes the last stage, followed by the CCFF. Then, the uncertainty-minimal query selection selects a fixed number of encoder features to serve as initial object queries for the decoder. Finally, the decoder with auxiliary prediction heads iteratively optimizes object queries to generate categories and boxes. The first output of the decoder layer is selected as the content query to refine by SACL. (b) Fusion module in CCFF, which contains two 1 × 1 convolutions and N RepBlocks composed of RepConv [61]. (c) GSAF module, which contains two 1 × 1 convolutions, one GSA block, and two PConvs.

strategy, which balances the problem of generating false-negative samples when anchors define positive and negative samples. Subsequently, a hard sample learning module is proposed to mine the potential hard negative samples in the sample data and focus on the optimization of the hard samples, which successfully and efficiently mines the hard samples and performs focused learning. Notably, there has been research on unsupervised object discovery. Wang et al. [57] proposed a methodology geared toward identifying salient objects without relying on manual annotations, which incorporates weakly supervised contrastive learning (WCL) to enhance semantic information exploration. Additionally, Li et al. [58] proposed the multiclass min-margin contrastive learning (MMCL) mechanism, which addresses that the distribution of content queries of deformable DETR-based models has no apparent class boundaries.

In summary, while contrastive learning has been widely adopted in CNN-based models with promising results, DETR-based methods still lack effective approaches. Actually, though the content queries are initialized in various ways [19] [59] [60], they all serve as priors for semantic content. To fully exploit the semantic information in content queries, a supervised contrastive learning strategy, the SACL strategy, is proposed, which helps decouple foreground and background features through the prior knowledge in the content queries.

III. Method

This section introduces the proposed S²Net for RSOD. The overall structure is illustrated in Fig. 2. RT-DETR is selected as the benchmark framework due to its superior performance in maintaining both high-accuracy and end-to-end real-time capability for multiscale object detection tasks. To address the challenges posed by significant variations in objects within RSIs, three novel methods are designed for different stages of the baseline to improve the multiscale object detection performance. GSAF is proposed for enhanced spatial context modeling and multiscale feature fusion, ER-IoU is employed to improve localization accuracy, and SACL is designed to strengthen query embedding discrimination. Details could be found in Sections III-A–III-C.

A. GSA-Based Fusion

RSIs involve scale variations and complex spatial layouts, challenging conventional feature fusion designs. Consequently, to balance the focus on all scales and preserve fine-grained details, GSAF is designed, which integrates a GSA block and two PConv modules, shown in Fig. 2(c). GSA models long-range spatial dependencies and spatial contextual patterns, while PConv enhances local structural awareness. Their synergy helps handle multiscale features and yields rich spatial representations.

As shown in Fig. 3, GSA is designed to capture all scales of features and enhance spatial context modeling while maintaining efficiency. Input features are divided into groups to promote diverse pattern learning through specialized processing. These grouped features are passed into a dual-branch parallel. The 1 × 1 branch aggregates spatial information

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 11:14:37 UTC from IEEE Xplore. Restrictions apply.

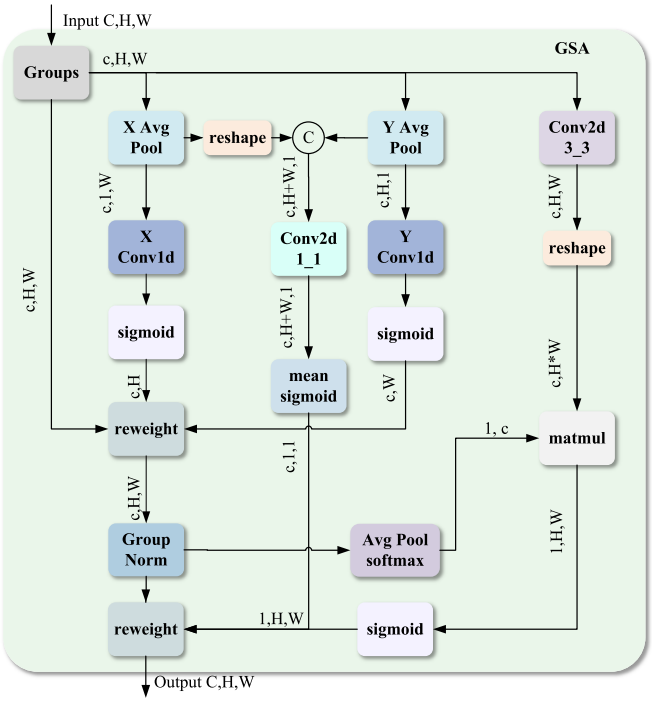


Fig. 3. Details of GSA. c = C G, where c denotes the number of grouped channel, C denotes the input and output channel, and G denotes the number of groups.

along horizontal and vertical axes via adaptive pooling, capturing long-range dependencies and encoding global context. This branch then splits into two complementary paths: one fuses height and width cues for global attention weighting, while the other applies 1-D convolutions to encode two routes and aggregates them via multiplication. In parallel, the 3×3 branch collects multiscale spatial cues through depthwise convolutions and large local receptive fields.

A cross-branch interaction is introduced by modulating the 3×3 features using attention weights derived from the 1×1 branch via global pooling and softmax operations, followed by a sigmoid gating. This adaptive fusion enables GSA to further enlarge the feature space and encode the global information, significantly enhancing feature representation for multiscale RSOD.

Complementing GSA's global modeling, the PConv [24] modules focus on enhancing local structural awareness. Its core step is formulated as follows:

$$Y_0 = \text{Conv}\left(c_1, \frac{c_2}{4}, (1,k), s\right)(\text{ZeroPad2d}(k,0,1,0)(X)) \quad (1)$$

where X denotes the input tensor; c_1 and c_2 denote the input and output channels, while k and s denote the kernel size and the stride. The other three will pad with (0, k, 0, 1), (0, k, 0, 1), and (0, k, 0, 1), with kernels (1, k), (k, 1), and (k, 1). Then 4 different convolution output features Y_i will be concatenated and projected to the output.

With its pinwheel-shaped padding design, PConv achieves an efficient receptive field expansion with minimal parameter increase compared to normal convolution, which is beneficial for collecting local spatial features in RSI. To strike a balance between performance and efficiency, two consecutive PConv (4, 3) modules are adopted in GSAF. In particular, a PConv kernel length of 4 in the first layer provides a more effective receptive field, crucial for capturing the features. As the feature map and target size reduce during downsampling, a kernel length of 3 is sufficient for subsequent layers.

By combining GSA and PConv within GSAF, the module effectively unifies global spatial context with local cues. While GSA strengthens the representation of long-range dependencies, PConv complements it by refining fine-grained structural patterns. This cooperation allows the fused features to remain both globally consistent and locally discriminative, which is crucial for detecting remote sensing objects of varying sizes and shapes. Moreover, the total number of parameters and computational cost of the proposed GSAF module are reduced compared to the fusion module of the benchmark framework.

In the model design of CCFF, for efficiency and performance considerations, two ffusion modules are used in the top-to-down FPN stream, while two proposed GSAF modules are used in the down-to-top PAN stream. Based on its robust and spatial–context-aware feature representation, CCFF with GSAF works in conjunction with attention-based intrascale feature interaction (AIFI), which leverages self-attention to model global spatial dependencies and refine intertoken relationships across the entire image, thereby enhancing the overall representational capacity for multiscale remote sensing scenes.

B. Edge-Ratio IoU

To evaluate localization quality in object detection, an advanced bounding box regression loss function named ER-IoU is proposed, providing more robust and precise training signals. Built upon the edge-alignment mechanism, ER-IoU introduces a ratio-aware regularization term that penalizes aspect ratio mismatches between predicted boxes and ground truths. This encourages both boundary-level precision and shape consistency, leading to improved localization accuracy. The ER-IoU loss is formulated as follows:

$$L_{\text{ER-IoU}} = L_{\text{IoU}} + R_{\text{edge}} + \gamma * R_{\text{ratio}} \quad (2)$$

where L_{IoU} denotes the IoU loss, R_{edge} denotes the main edge-alignment regression, and R_{ratio} is an aspect ratio penalty term. γ is a modulation factor of R_{ratio}, formulated as follows:

$$\gamma = \frac{R_{\text{ratio}}}{L_{\text{IoU}} + R_{\text{ratio}}} \quad (3)$$

where γ is introduced to balance the contribution of R_{ratio} to the overall loss. In the early stage of the regression process, when L_{IoU} is high, improving the location and overlap should take priority, and γ is reduced to avoid premature focus on shape alignment. Conversely, when it comes to a low L_{IoU} indicating sufficient overlap, γ increases to emphasize the role of the aspect ratio penalty term R_{ratio}. L_{IoU} notes the overlap between the predicted box and ground-truth box, formulated as follows:

$$L_{\text{IoU}} = 1 - \text{IoU}. \quad (4)$$

In the training process, R_{edge} directly minimizes the positional offsets between the corresponding edges of the anchor

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 11:14:37 UTC from IEEE Xplore. Restrictions apply.

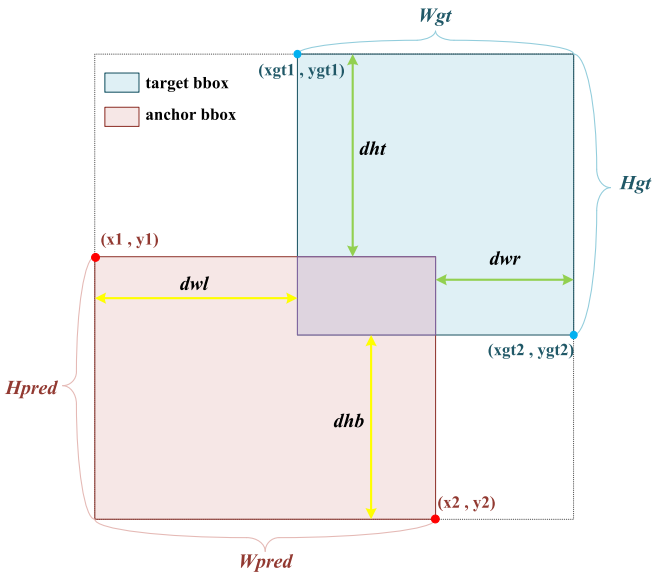


Fig. 4. ER-IoU loss, using only the edge lengths of the target box as the denominator in the loss factor.

and target boxes, avoiding unnecessary enlargement of the area of the anchor box, which is calculated as follows:

$$R_{\text{edge}} = 1 - e^{-O^2} \tag{5}$$

where O denotes the normalized edge offset between the predicted box and ground-truth box. It is calculated as as follows:

$$O = \left(\frac{dw_l}{W_{\text{gt}}} + \frac{dw_r}{W_{\text{gt}}} + \frac{dh_t}{H_{\text{gt}}} + \frac{dh_b}{H_{\text{gt}}}\right) \Big/ 4 \tag{6}$$

where dw_l, dw_r, dh_t, and dh_b are the absolute values of the distances between the corresponding edges of the predicted anchor box and target box, and W_{gt} and H_{gt} represent the width and height of the target box, as illustrated in Fig. 4.

Furthermore, an additional penalty term R_{ratio} is introduced to explicitly model the deviation in width–height proportions, which is formulated as follows:

$$R_{ratio} = \tanh^2\left(\log\left(\frac{r_{\text{gt}}}{r}\right)\right) \tag{7}$$

where r_{gt} and r denote the aspect ratio, formulated as follows:

$$r = \left|\frac{W_{\text{pred}}}{H_{\text{pred}}}\right|, \quad r_{\text{gt}} = \left|\frac{W_{\text{gt}}}{H_{\text{gt}}}\right| \tag{8}$$

where W_{pred}, H_{pred}, W_{gt}, and H_{gt} represent the width and height of the anchor box and the target box.

The formulation (7) offers several advantages: first, the logarithmic term $\log(r_{gt}/r)$ quantifies the proportional discrepancy between the predicted ratio r and the ground-truth ratio r_{gt}. It naturally ensures symmetry, since both overestimation and underestimation are penalized equally. Moreover, it is scale-invariant, as only the ratio matters, not the absolute width or height. Additionally, it is sensitive to small ratio deviations while remaining smooth across a wide range. By applying the $\tanh^2$ nonlinearity, as shown in Fig. 5, the penalty is smoothly mapped to the interval $[0, 1)$. Importantly, the curvature of $\tanh^2$ makes the penalty most sensitive when $\log(r_{gt}/r)$ is near zero (i.e., when r is close to r_{gt}). This design implicitly emphasizes more precise adjustments in medium-to-high IoU samples, which are critical for fine-grained localization, while saturating for extreme outliers to avoid distracting the optimization.

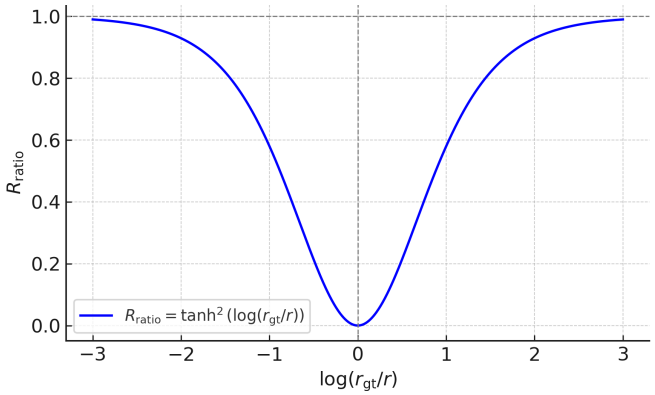


Fig. 5. Curve showing the variation of R_{ratio} with $\log(r_{\text{gt}}/r)$.

Moreover, the dynamic adjustment of the adaptive factor γ allows smooth optimization from coarse to fine alignment. Through this joint mechanism, the model ultimately improves localization precision.

To further analyze the efficiency of ER-IoU, the gradient of R_{edge} with respect to box coordinate x_1 is calculated using chain rule as follows:

$$\frac{\partial R_{\text{edge}}}{\partial x_1} = 2Oe^{-O^2} \cdot \frac{1}{4|W_{\text{gt}}|} \cdot \text{sgn}\left(x_1 - x_{gt1}\right) \tag{9}$$

where $\text{sgn}(\cdot)$ denotes the sign function, x_1 denotes one of the anchor box coordinates, and x_{gt1} denotes one of the target box coordinates, as shown in Fig. 4. Similar derivations can be obtained for x_2, y_1, and y_2. Thus when edges are perfectly aligned, $O = 0$, and the gradient vanishes—loss is minimized. As the offset increases, O increases but e^{-O^2} decreases, leading to strong correctional gradients. This ensures that the predicted box is explicitly encouraged to match the ground-truth boundaries, avoiding undesirable enlargement behaviors seen in GIoU.

C. Semantic-Adaptive Contrastive Learning

To offer direct supervision to the evolving query embeddings, the SACL is proposed—a novel auxiliary strategy that explicitly enhances the discriminative power of decoder query embeddings at intermediate layers, as illustrated in Fig. 6. SACL introduces an overall loss (L_{SAC}) consisting of two parts: interclass separation loss (L_{ICS}) and ICC loss (L_{ICC}), which respectively group and cluster queries, potentially breaking the limit of the learned representations

$$L_{\text{SAC}} = \alpha L_{\text{ICS}} + \beta L_{\text{ICC}} \tag{10}$$

where α and β are scalar weights that balance the contributions of the two components.

1) *Interclass separation (ICS):* Query embeddings from different classes should be well separated in the feature space.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 11:14:37 UTC from IEEE Xplore. Restrictions apply.

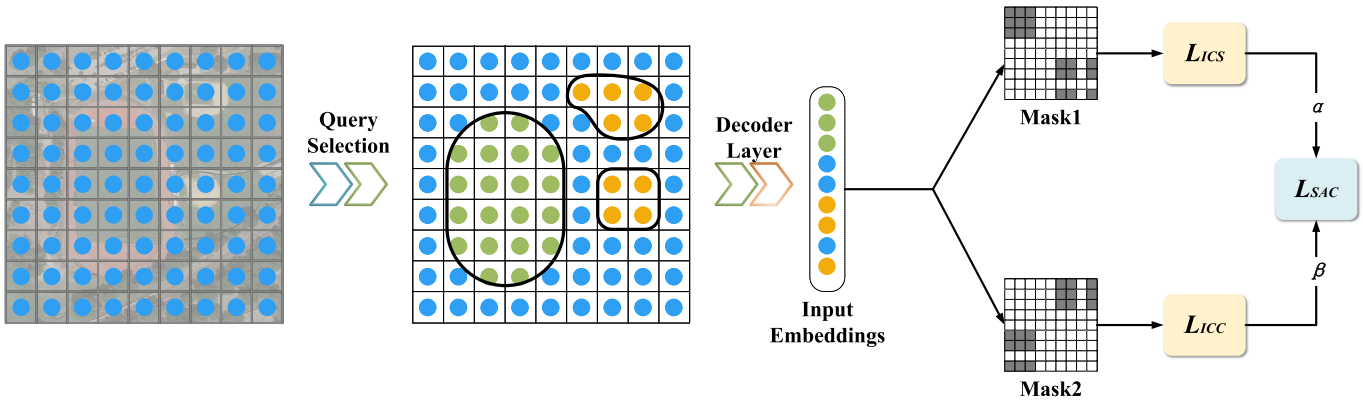


Fig. 6. Working mechanism of SACL. The input feature undergoes a query selection mechanism and is processed by the first decoder layer to output content embeddings. Two dynamic masks are generated for these embeddings based on the ground-truth labels, and the losses L_{ICS} and L_{ICC} are computed separately, then weighted to obtain the final L_{SAC}.

2) *Intra-Class Compactness (ICC):* Embeddings belonging to the same class should be clustered tightly.

When optimized jointly, these two components form a complementary mechanism: L_{ICS} mitigates category overlap by enlarging interclass distances, while L_{ICC} moderates scattered clusters by tightening intraclass distributions. To strike a balance between them effectively creates a discriminative embedding space where queries are simultaneously well-separated across classes and internally consistent within classes. Consequently, a parameter analysis of α and β has been conducted in Section IV-C. Such a dual constraint is crucial for decoder queries, as it mitigates semantic confusion under class imbalance and ensures robust embeddings for downstream classification tasks in RSOD.

To encourage better class-level discrimination, an interclass separation loss that penalizes high similarity between embeddings from different target categories is defined as follows:

$$L_{\mathrm{ICS}} = \frac{-1}{\sum M1_{i,j}} \sum_{i=1}^{M1} \sum_{j=1}^{M1} M1_{i,j} \log\left(1 - \mathrm{sim}\left(h_i, h_j\right)\right) \tag{11}$$

where h_i and h_j denote two decoder query embeddings in the same batch, and sim(.) is the cosine similarity function. $M1_{i,j}$ is a binary mask, given as follows:

$$M1_{i,j} = \begin{cases} 1, & \mathrm{target}_i \neq \mathrm{target}_j \\ 0, & \text{otherwise.} \end{cases} \tag{12}$$

where $\mathrm{targets}_i$ and $\mathrm{targets}_j$ identify negative pairs (i.e., embeddings from different object categories). These two formulations minimize the similarity between negative pairs, encouraging interclass boundaries to be preserved in the embedding space.

To simultaneously ensure that embeddings within the same category remain close, an ICC loss that maximizes similarity among positive pairs is defined as follows:

$$L_{\mathrm{ICC}} = \frac{-1}{\sum M2_{i,j}} \sum_{i=1}^{M2} \sum_{j=1}^{M2} M2_{i,j} w_{i,j} \log\left(\mathrm{sim}\left(h_i, h_j\right)\right) \tag{13}$$

where h_i and h_j denote two decoder query embeddings in the same batch, and *sim(.)* is the cosine similarity function. $w_{i,j}$ is an optional confidence-based weight, given as follows:

$$w_{i,j} = \exp\left(-c \cdot \mathrm{rank}\left(h_i, h_j\right)\right) \tag{14}$$

where c denotes the constant and rank denotes the similarity rank, which can further modulate each pair's contribution. And $M2_{i,j}$ denotes a second binary mask, given as follows:

$$M2_{i,j} = \begin{cases} 1, & \mathrm{target}_i = \mathrm{target}_j \neq \text{background} \\ 0, & \text{otherwise} \end{cases} \tag{15}$$

where $\mathrm{targets}_i$ and $\mathrm{targets}_j$ denote positive pairs (i.e., embeddings assigned to the same nonbackground category).

As stated above, SACL leverages a supervised masking mechanism to dynamically select valid similarity pairs. By comparing target labels within each batch, a binary mask enforces positive/negative pair selection based on class identity and excludes background-class embeddings from optimization, as their interclass similarity is ill-defined. In fact, constructing the masks based on real samples is based on the core design of supervised contrastive learning, where ground-truth labels are used to distinguish positive and negative pairs. This adaptive mechanism ensures that SACL can effectively handle the weak semantic activations and class imbalance issues across datasets frequently encountered in RSOD, improving detection accuracy even under cluttered and distorted visual conditions.

IV. Experiment

A. Datasets and Implementation Details

To comprehensively evaluate the effectiveness and generalization ability of the proposed S^2Net, three remote sensing datasets are used: NWPU VHR-10 [21], DIOR [22], and RSOD [23]. These datasets collectively cover both small- and large-scale detection tasks, include objects with diverse spatial resolutions, and present challenges such as dense object distribution, high intraclass variation, and cross-scene variability.

1) *NWPU VHR-10 [21]:* The NWPU VHR-10 dataset is a publicly available ten-class geospatial object detection dataset used for research purposes only. These ten

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 11:14:37 UTC from IEEE Xplore. Restrictions apply.

classes of objects are airplane (AL), ship (SH), storage tank (ST), baseball diamond (BD), tennis court (TC), basketball court (BC), ground track field (GF), harbor (HB), bridge (BR), and vehicle (VH). This dataset contains very-high-resolution (VHR) RSIs that were cropped from Google Earth and Vaihingen dataset and then manually annotated by experts. Specifically, it contains 800 images, including 715 with spatial resolutions ranging from 0.5 to 2 m and 85 with a resolution of 0.08 m, among which 650 are annotated with object instances. In total, 80% of the 650 images are randomly divided as the training set and 20% as the testing set.

2) *DIOR [22]:* The DIOR dataset is a large-scale benchmark dataset for object detection in optical RSIs, which consists of 23 463 images and 192 518 object instances annotated with horizontal bounding boxes. It covers 20 object classes: airplane (AL), airport (AT), baseball field (BF), BC, bridge (BG), chimney (CM), dam (DM), expressway service area (EA), expressway toll station (ES), golf course (GC), GF, harbor (HB), overpass (OP), ship (SP), stadium (SD), ST, TC, train station (TS), vehicle (VH), and wind mill (WM). The images in the dataset were sourced from Google Earth, with a resolution of 800 × 800 and a spatial resolution ranging from 0.5 to 30 m. It is large-scale, encompassing a substantial number of images, object categories, and annotated instances. It features significant variability in object sizes, not only due to differences in spatial resolution but also stemming from interclass and intraclass scale diversity. Additionally, the dataset includes images captured under a wide range of imaging conditions—such as varying weather, seasons, and sensor qualities—introducing rich environmental complexity. Furthermore, the high degree of interclass visual similarity and intraclass appearance diversity poses additional challenges for robust object detection models. This dataset is divided into 11 725 images (50% of the dataset) as the training set and the remaining 11 738 images (50% of the dataset) as the testing set.

3) *RSOD [23]:* The RSOD dataset is an open-access resource supporting object detection in remote sensing imagery, presented by Wuhan University in 2017. It covers four object categories, including 4993 aircraft, 1586 oil tanks, 191 playgrounds, and 180 overpasses in 2326 RSIs, with instances exhibiting substantial size diversity. The unified strategy in [62] is adopted to randomly divide 75% for training and 25% for testing.

The experimental environment is the PyTorch framework (PyTorch2.0.1) in the Ubuntu 20.04 operating system, and all experiments are performed on a single NVIDIA RTX A6000 GPU with 48 GB memory. ResNet-18 pretrained on ImageNet is adopted as the backbone. In the training phase, comprehensive training is conducted with differentiated schedules: 200 epochs on both NWPU VHR-10 and RSOD, and 160 epochs on DIOR. For the loss function, the weight of the bounding box regression loss is 5.0, the classification loss weight is 1.0, the ER-IoU loss weight is 2.0, and the L_{SAC} weight is 1.0. The AdamW optimizer is employed with a learning rate of 0.0001, $\beta_1 = 0.9$, $\beta_2 = 0.999$, and a weight decay of 0.0001. To prevent over-regularization of normalization layers, weight decay is disabled for parameters matching normalization-related patterns. The linear warm-up strategy is also implemented. The first quarter of iterations per epoch is designated as the warm-up steps, with a warm-up start ratio of 0.001 following the setup of the baseline. The size of the input images for the network is 640 × 640 for three datasets. The batch size is 16 for training while 32 for testing.

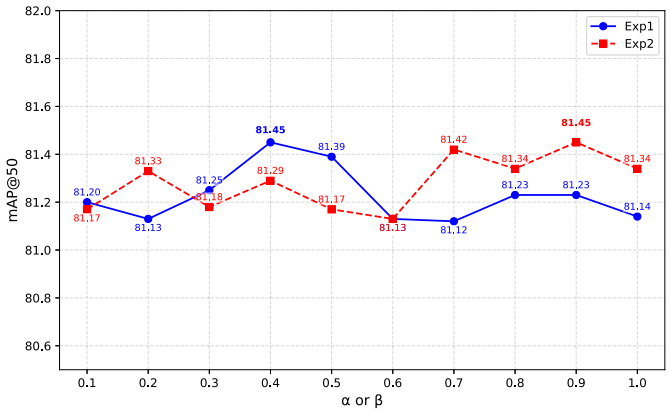


Fig. 7. Detection performance for different values of α and β on DIOR.

B. Evaluation Metrics

This section details the evaluation framework for assessing model performance. First, the confusion matrix is introduced to evaluate the accuracy. It consists of four parts, which are true positive (TP), true negative (TN), false positive (FP), and FN. If a classifier predicts a sample as positive and it is positive, it is called a TP. On the other hand, if the classifier predicts a sample as positive but it is actually negative, it is called an FP. When the classifier correctly predicts that a sample is negative, it is called a TN. However, if the classifier predicts a sample as negative but it is actually positive, it is called an FN. Precision and recall are expressed as follows:

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}}. \tag{16}$$

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}}. \tag{17}$$

In object detection tasks, boxes that overlap with a certain amount of similarity, known as the IoU score, are classified as TP instances or FP instances. FN instances occur when the model fails to detect a ground-truth bounding box. The AP value changes with different IoU thresholds, with a higher IoU threshold resulting in a lower AP value. Higher AP indicates better performance at a given IoU threshold. The object detection evaluation indexes include precision, recall, and mean average precision (mAP). The accuracy calculation formula is as follows:

$$\text{AP} = \int_0^1 \text{P}(R)\, dR. \tag{18}$$

$$\text{mAP} = \frac{1}{N_{\text{cls}}} \sum_{i=1}^{N_{\text{cls}}} \text{AP}_i \tag{19}$$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 11:14:37 UTC from IEEE Xplore. Restrictions apply.

TABLE I
COMPARISON OF DIFFERENT METHODS ON NWPU VHR-10. THE BEST RESULTS ARE MARKED IN BOLD

Method	AL	SH	ST	BD	TC	BC	GT	HB	BR	VH	mAP
Faster R-CNN [9]	99.0	91.3	97.3	97.0	92.6	91.1	99.3	92.3	86.9	92.9	94.0
ReDet [63]	99.9	93.5	98.9	94.3	95.5	95.5	100	88.1	70.9	**96.1**	93.3
DINO [59]	96.9	89.6	93.9	96.4	90.6	89.2	94.6	91.3	86.9	89.8	91.9
SuperYOLO [64]	99.3	86.0	99.0	97.5	97.5	77.5	99.3	93.6	86.2	91.3	92.7
ViTAE-B [65]	**100**	87.8	97.8	96.4	94.5	97.5	**100**	86.9	89.5	96.0	94.6
LSK-Net [66]	99.0	**95.4**	96.1	97.8	94.6	96.3	99.7	88.6	94.4	92.9	95.5
KLDet [67]	**100**	92.3	96.5	95.3	94.7	91.3	99.7	82.3	70.9	93.4	91.6
BSSOCNet [68]	99.9	85.9	89.7	97.3	96.6	98.8	**100**	**97.7**	85.4	90.8	94.2
RT-DETR [20]	97.3	93.6	**99.8**	94.8	94.2	98.7	99.9	97.3	75.6	89.9	94.1
CFIMNet [69]	**100**	86.7	92.5	96.3	91.4	95.8	**100**	97.1	**95.0**	93.2	94.8
SED-DETR [70]	99.9	92.8	97.9	95.9	94.9	93.0	94.5	94.6	94.3	89.5	94.7
TBNet [71]	**100**	95.1	97.4	97.4	**97.6**	99.0	**100**	94.6	80.8	92.7	95.4
S^2Net (ours)	94.2	93.9	99.6	**97.8**	95.8	**99.3**	**100**	**97.7**	92.3	92.4	**96.3**

TABLE II
COMPARISON OF DIFFERENT METHODS ON DIOR. THE BEST AND SUBOPTIMAL RESULTS ARE MARKED IN RED AND BLUE, RESPECTIVELY

Method	AL	AT	BF	BC	BG	CM	DM	EA	ES	GC	GF	HB	OP	SP	SD	ST	TC	TS	VH	WM	mAP
Faster R-CNN [9]	54.9	81.4	70.1	88.5	39.3	79.3	60.6	77.8	59.8	78.2	83.1	52.9	57.1	68.4	74.5	52.6	84.6	59.7	36.0	78.9	66.9
RetinaNet [72]	61.5	70.3	76.0	86.5	40.0	75.6	61.6	75.8	68.2	77.4	78.1	36.1	54.3	73.0	59.1	61.8	87.4	39.3	43.3	86.2	65.5
ABNet [73]	66.8	84.0	74.9	87.7	50.3	78.2	67.8	85.9	74.2	79.7	81.2	55.4	61.6	75.1	74.0	66.7	87.0	62.2	53.6	89.1	72.8
GLSANet [38]	95.8	78.9	92.9	87.9	50.7	81.1	55.5	79.8	74.1	71.6	87.6	66.4	65.5	95.2	92.4	86.3	94.8	50.6	62.1	89.2	77.9
PANet [74]	81.4	82.7	83.4	87.1	43.0	78.2	69.2	84.0	67.1	78.8	82.5	64.6	62.9	90.9	79.9	65.9	91.5	66.0	52.3	87.5	74.9
CoF-Net [75]	84.0	85.3	82.6	90.0	47.1	80.7	73.3	89.3	74.0	84.5	83.2	57.4	62.6	82.9	77.6	68.2	89.9	68.7	49.3	85.2	75.8
BSSONCNet [68]	87.4	78.4	74.5	89.1	44.3	77.6	59.8	58.5	60.2	78.2	71.8	62.4	59.2	89.7	65.7	74.0	88.5	55.6	54.4	84.8	70.7
CFIMNet [69]	72.5	74.1	79.0	89.8	45.6	79.1	67.7	82.3	77.6	80.7	81.6	38.4	59.5	75.9	67.4	68.6	89.9	51.8	52.7	89.6	71.2
KLDet [67]	67.5	81.1	75.5	88.2	47.4	78.6	65.0	82.9	72.7	79.9	80.8	57.1	61.2	91.2	74.6	75.2	87.4	56.9	56.0	90.2	73.5
SFSANet [76]	87.8	83.3	87.8	86.7	48.8	78.5	68.2	85.0	76.5	73.2	78.6	62.9	63.6	90.2	83.4	79.4	83.8	61.5	67.9	84.7	76.6
AFANet [77]	92.3	86.2	85.8	92.6	49.6	79.7	68.8	91.5	78.3	84.6	85.8	66.8	65.1	76.4	88.7	71.4	93.6	66.8	56.5	87.3	78.4
AFSNet [78]	91.6	83.4	96.1	94.9	56.2	87.6	79.4	64.6	76.8	87.9	73.3	60.5	61.7	94.3	87.4	77.0	89.2	65.7	57.9	92.9	78.9
RT-DETR [20]	92.3	89.8	89.1	90.8	58.6	81.6	76.1	91.8	82.8	83.3	87.2	68.0	68.2	78.5	82.2	78.2	92.6	73.2	61.1	92.9	80.9
ViT-L + RVSA w MTP [79]	93.8	91.6	87.7	92.1	64.6	85.9	81.4	94.3	84.5	87.3	86.7	64.5	72.0	76.3	85.6	62.6	92.5	79.7	47.6	90.2	81.1
TBNet [71]	64.9	86.8	76.6	89.2	50.6	80.0	74.3	86.4	74.7	82.5	83.6	56.4	64.9	79.1	77.8	58.1	88.2	69.4	44.6	87.0	73.6
STDet [80]	72.6	91.0	82.8	89.5	53.9	80.2	77.6	89.3	79.8	85.8	87.1	51.6	65.4	76.1	82.1	71.0	89.2	75.1	58.0	92.2	77.5
UniconDet [27]	89.7	85.8	86.7	90.5	52.9	79.3	69.2	82.6	80.2	80.9	83.8	59.1	63.2	87.5	76.8	80.8	90.1	65.2	60.8	87.0	77.6
Relation-DETR [81] (+PDQR) [82]	84.7	92.1	83.7	91.5	57.9	86.5	78.2	92.7	81.1	86.0	87.6	69.1	68.7	80.0	84.3	78.1	92.0	71.1	62.7	92.3	81.0
S^2Net(ours)	92.7	89.9	90.1	91.0	58.5	82.1	76.3	93.1	84.4	84.6	87.5	67.6	68.6	79.1	83.4	78.5	92.9	74.5	61.0	93.3	81.4

where P and R refer to precision and recall, and N_{cls} represents the total number of categories. Notably, in comparison experiments, mAP is calculated as the mean of AP_{50} for each category. In ablation experiments and parameter analysis, following the MS COCO benchmark, $AP_{50:95}$ and AP_{50} are used to evaluate the overall accuracy of all categories.

Second, the computational complexity and the model parameters are also evaluated. The number of parameters is presented in millions (M) for clarity. The computational

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 11:14:37 UTC from IEEE Xplore. Restrictions apply.

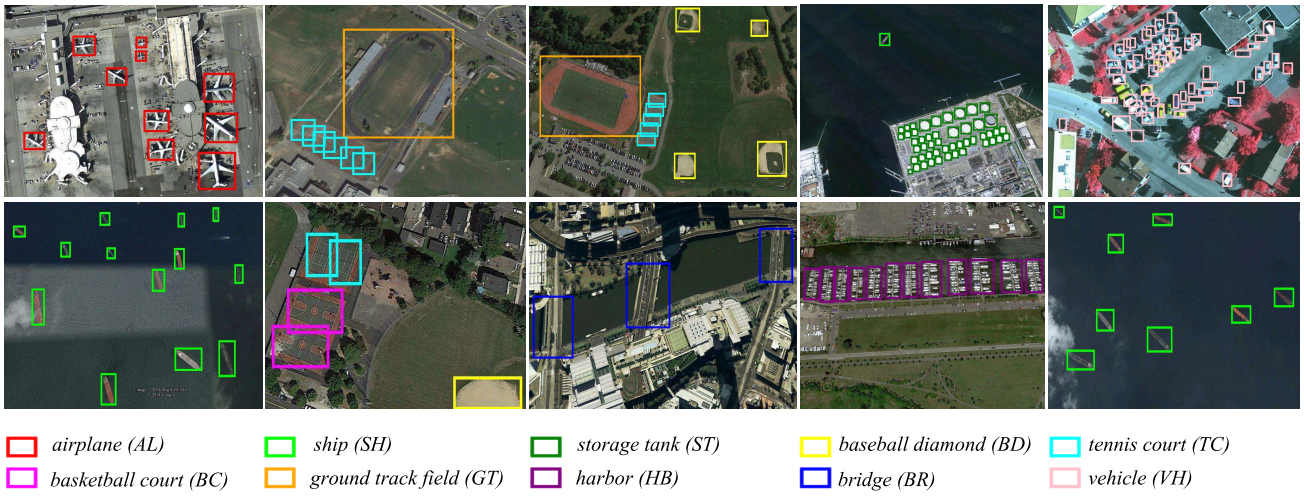


Fig. 8. Some representative predictions of S²Net on NWPU VHR-10 dataset.

complexity is assessed using GFLOPs, which quantify the number of floating-point operations

$$\text{Parameters}\,(M) = \frac{\text{Para}}{10^6} \tag{20}$$

$$\text{GFLOPs} = \frac{F}{10^9} \tag{21}$$

where Para is the total number of parameters, and F is the total number of floating-point operations.

C. Parameter Analysis

In SACL, α and β control the contributions of L_{ICS} and L_{ICC} during training. To investigate the impact of the hyper-parameters α and β on the overall performance of SACL, we conducted a systematic grid search on the DIOR dataset. As illustrated in Fig. 7, control variables method is used to obtain reliable parameter settings. For Exp1, the value of β is fixed at 0.9 while α varies from 0.1 to 1.0. For Exp2, the value of α is fixed at 0.4 while β varies from 0.1 to 1.0. The results reveal that the best performance of AP_{50} occurs when $\alpha = 0.4$ and $\beta = 0.9$, achieving 96.3% on DIOR. Notably, when the parameters slightly deviate around this optimal point, S²Net also maintained consistently strong performance in other evaluation metrics. This indicates the robustness and stability of the proposed SACL strategy.

Furthermore, we transferred the optimal parameter setting to two additional remote sensing datasets, NWPU VHR-10 and RSOD, where it achieves 96.3% and 96.8% of mAP_{50}, respectively. As shown in the comparison experiments in Section IV-D, this configuration consistently outperformed several advanced and representative models, demonstrating the strong generalization ability and robustness of our proposed strategy across diverse datasets.

D. Comparing State-of-the-Art Methods

1) Results on NWPU VHR-10: With the results presented in Table I, the experiments are conducted on the NWPU VHR-10 dataset. S²Net is benchmarked against several representative and novel methods, including Faster R-CNN [9], ReDet [63], DINO [59], SuperYOLO [64], ViTAE-B [65], LSK-Net [66], KLDet [67], BSSOCNet [68], RT-DETR [20], CFIMNet [69], SED-DETR [70], and TBNet [71]. Among these models, S²Net achieves the best performance with 96.3% mAP. In particular, compared to some latest methods KLDet, SED-DETR, and TBNet, S²Net has improvements of 4.7%, 1.6%, and 0.9%, respectively. Furthermore, S²Net achieves the best accuracy in 4 out of 10 categories, with 97.8% on BD, 99.3% on BC, 100% on ground track field, and 97.7% on harbor. Representative detections are visualized in Fig. 8. It can be seen that the consistent and balanced performance achieved by S²Net across all dataset categories (with all the values higher than 92.3%) provides strong evidence for its robustness in handling multiscale data distributions, making it competitively stand out from other models.

2) Results on DIOR: The proposed S²Net is compared with several detection models, including general methods such as Faster R-CNN [9] and RetinaNet [72], and advanced methods such as ABNet [73], GLSANet [38], PANet [74], CoF-Net [75], BSSONCNet [68], CFIMNet [69], KLDNet [67], SFSANet [76], AFANet [77], AFSNet [78], RT-DETR [20], MTP [79], TBNet [71], STDet [80], UniconDet [27], and Relation-DETR [81] (+PQDR [82]). The concrete results are presented in Table II. Net achieves the state-of-the-art result with 81.4% mAP. It outperforms some of these latest methods like TBNet, STDNet, UniconDet, and PDQR, showing enhancements of 7.8%, 3.9%, 3.8%, and 0.4%, respectively. In detail, the model achieves the suboptimal performance with 93.1% on expressway-service-area, 84.4% on expressway-toll-station, and 87.5% on ground track field, and the best with 93.3% on wind mill, demonstrating its superior ability to detect diverse target types. Comparison visualization results of 3 advanced methods are shown in Fig. 9. It can be seen that, when the targets in the image exhibit a wide range of scales, S²Net alleviates the negative impact between objects of different sizes, reducing the likelihood of focusing solely on a single scale, as observed in scenarios such as ship versus harbor and vehicle versus TC. Moreover, in scenes like bridge and harbor, S²Net helps mitigate issues such as large targets being fragmented and small targets being missed (i.e., FNs).

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 11:14:37 UTC from IEEE Xplore. Restrictions apply.

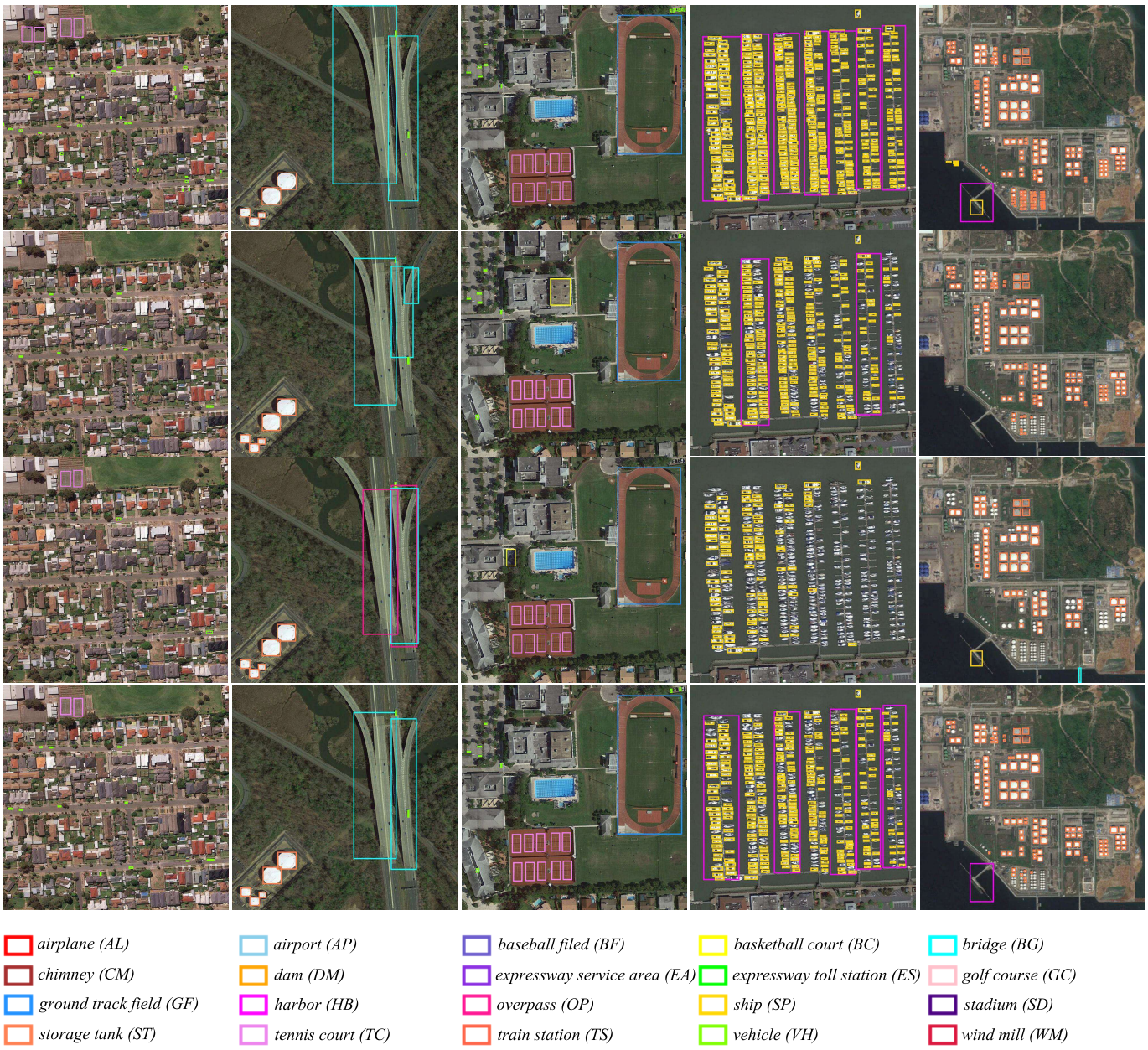


Fig. 9. Comparison visualization results of different methods on the DIOR dataset. Top to bottom: Ground Truth, RT-DETR, ViT-L + RVSA (w MTP), and S^{2}Net.

In terms of accuracy, Net reduces both class misclassification (where targets are assigned the wrong category) and FP detections (where background areas are misidentified as targets). Overall, S^{2}Net effectively balances multiscale object detection, thereby improving detection precision, reducing false alarm rates, and increasing the stability of the model under different conditions.

3) Results on RSOD: As shown in Table III, comparative experiments are also conducted on RSOD dataset. Several representative methods are compared with S^{2}Net, including Faster R-CNN [9], CANet [62], AGMF-Net [83], TMAFNet [84], Yang [85], AMFLW-YOLO [86], PANet [74], CoF-Net [75], GLSANet [38], SFSANet [76], SESA-Net [87], RT-DETR [20], Swin Transformer-421C [88], MSNet [89], PGCFNet [90], and SOD-YOLOv10 [91]. The proposed S^{2}Net exhibits the best performance on 3 out of 4 categories, with 99.0% on oil tank, 95.1% on overpass, 100% on playground. Furthermore, S^{2}Net obtained the best overall mAP with an accuracy of 96.8%, outperforming the above methods. Some representative detection results are exhibited in Fig. 10.

E. Ablation Study

In this section, an ablation experiment is performed to verify the effectiveness of GSAF, SACL, and ER-IoU in the proposed S^{2}Net. All experiments are evaluated on the NWPU VHR-10 test set. Table IV displays the results of ablation experiments.

1) Evaluation of GSAF: In Table IV, GSAF achieves 95.2% (↑ 0.6%) AP on NWPU VHR-10. This performance could be attributed to the advanced spatial modeling capabilities of GSAF, allowing it to preserve abundant low-level spatial

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 11:14:37 UTC from IEEE Xplore. Restrictions apply.



Fig. 10. Some representative predictions of S^{2}Net on the RSOD dataset.

TABLE III
COMPARISON OF DIFFERENT METHODS ON RSOD. THE BEST RESULTS ARE MARKED IN **BOLD**

Method	Aircraft	Oil tank	Overpass	Playground	mAP
Faster R-CNN [9]	81.3	96.8	85.5	95.5	89.8
CANet [62]	91.8	97.1	94.1	97.9	95.2
AGMF-Net [83]	96.0	**99.0**	82.4	99.7	94.3
TMAFNet [84]	96.2	98.9	85.2	99.8	94.5
Yang [85]	-	-	-	-	95.58
AMFLW-YOLO [86]	97.4	97.9	85.0	99.5	95.0
PANet [74]	**98.4**	98.7	90.7	97.9	96.4
CoF-Net [75]	96.5	98.8	85.5	99.3	95.0
GLSANet [38]	96.8	98.5	91.5	97.0	95.9
SFSANet [76]	96.0	97.1	88.4	98.7	95.0
SESA-Net [87]	95.7	97.9	90.2	96.8	95.2
RT-DETR [20]	91.2	97.7	93.4	99.6	95.5
Swin Transformer-421C [88]	95.9	-	-	-	95.8
MSNet [89]	96.4	97.3	90.9	98.0	95.6
PGCFNet [90]	-	-	-	-	95.8
SOD-YOLOv10 [91]	97.0	98.6	89.0	99.0	95.9
S^{2}Net (ours)	93.3	**99.0**	**95.1**	**100**	**96.8**

details during the multistage fusing. For a further and deeper analysis of GSAF and to demonstrate intuitive visualizations of its performance, we present the feature response heatmap from an intermediate layer. Specifically, we extract the feature maps from the pan\underlineblocks.0 and compute their squared L_2-norm across channels to generate a class-agnostic attention map, as shown in Fig. 11. By analyzing the results, the effectiveness of GSAF can be comprehensively interpreted from 3 aspects: target boundary clarity, structural integrity, and background suppression.

As evidenced in Fig. 11, the target response in the baseline model appears relatively blurred, with indistinct object contours and noticeable information diffusion at the edges. In contrast, heatmaps incorporating the GSAF module demonstrate significantly more continuous and well-defined object boundaries. This is particularly evident in features such as aircraft fuselages and wing edges, where the contours are markedly sharper. This kind of enhancement in boundary delineation primarily stems from the PConv modules. By leveraging their pinwheel-shaped, direction-sensitive design, PConv effectively captures orientation-specific edge information, accentuating object contours and fine-grained boundaries. Furthermore, the integration of shallow-level features through the initial 1×1 convolutions preserves rich low-level spatial

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 11:14:37 UTC from IEEE Xplore. Restrictions apply.

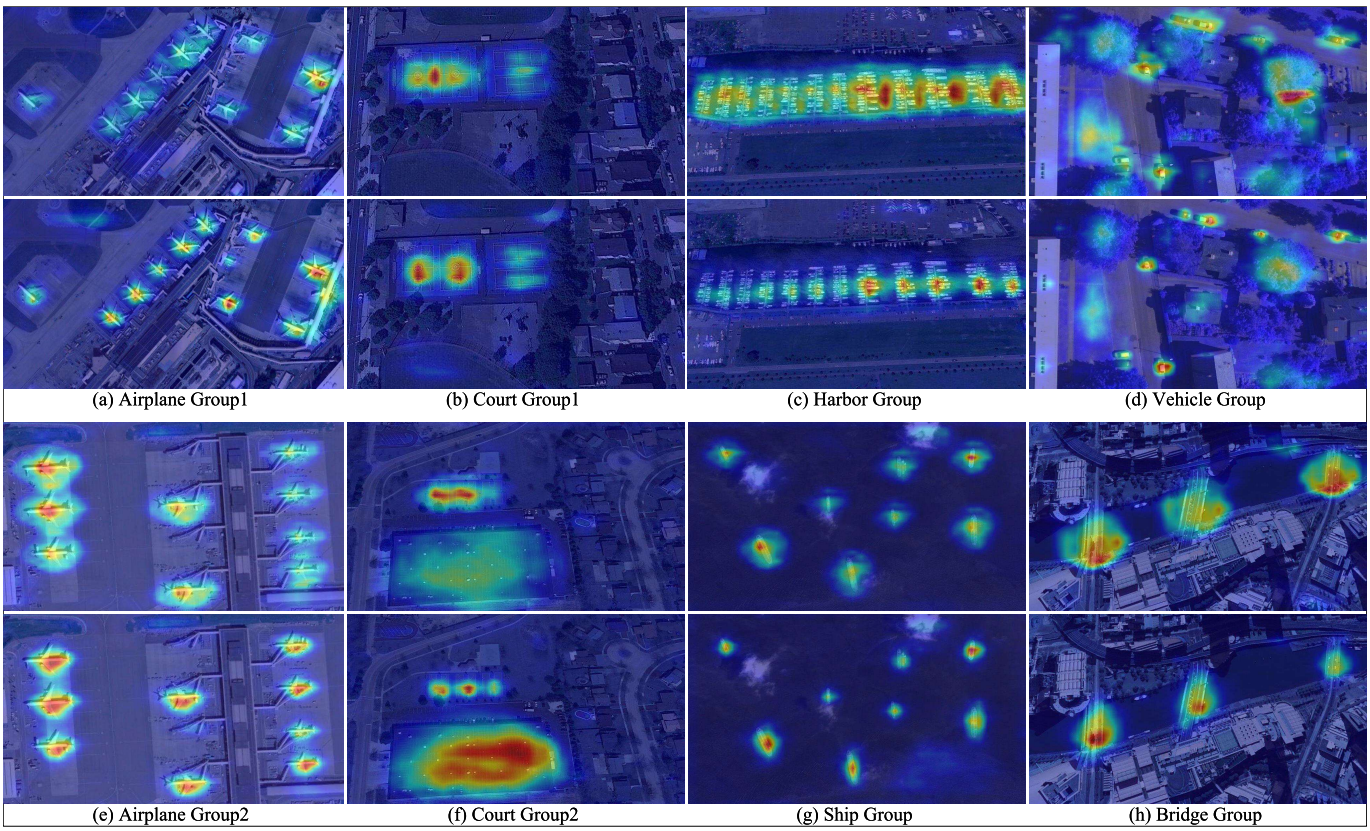


Fig. 11. Grouped CAM visualization results for different object categories. (a)–(h) Present groups of heatmaps illustrating airplanes, sports fields, as well as runways, harbors, vehicles, ships, and bridges. In each image group, the upper panel displays the heatmap generated by the baseline model, while the lower panel presents the heatmap obtained with GSAF augmentation. The NWPU VHR-10 dataset encompasses ten categories of remote sensing targets and the heatmap visualizations select representative examples covering the majority of these target classes.

details, which synergize with PConv's directional sensitivity. This combination enhances the model's ability to maintain sharp and accurate object edges, even under complex backgrounds or partial occlusions.

Meanwhile, the baseline detection results exhibit instability for multiscale objects, while the GSAF-enhanced outputs show adaptability. For example, in the scene of big and small ALs, the baseline results mainly focus on the big ones. Additionally, in the scene of three relatively small BCs and the big ground track field, it mainly focuses on the small ones. In contrast, it can be seen that the GSAF-enhanced outputs carry out awareness both on large and little objects. This accomplishment can be attributed to the flexible design of dual-branch parallel in GSA tailored to different scales, where the local receptive fields of neurons and the interaction enable the neurons to collect multiscale spatial information.

Otherwise, as observed in scenarios with complex target layouts (e.g., multiple adjacent courts, harbors, and streets), the GSAF-enhanced outputs exhibit a stronger focus on genuine target regions while significantly suppressing responses in surrounding background areas. For instance, baseline heatmaps display substantial artifacts at field boundaries and non-target locations, especially areas between multiple targets, whereas the GSAF-generated heatmaps demonstrate markedly reduced spurious activations. This improvement in background suppression results primarily from the global attention weighting pathway within GSA. By aggregating long-range dependencies through adaptive pooling and global attention weighting (softmax–sigmoid gating), GSA directs the model's focus toward true object regions while effectively suppressing activations in ambiguous or irrelevant background areas. This selective emphasis on target regions reduces false detections and enhances overall detection reliability, particularly in cluttered or densely populated environments. Collectively, the fine-grained spatial attention and directional feature modeling embedded in GSAF allow the model to allocate computational resources more precisely, improving both target localization and detection robustness under challenging remote sensing conditions.

2) Evaluation of ER-IoU: To evaluate the effectiveness of the proposed ER-IoU, the proposed method is extensively evaluated through both experiments on NWPU VHR-10 datasets and simulation-based analyses. As shown in Table IV, ER-IoU achieves 95.3% (↑ 0.7%) AP on NWPU VHR-10, outperforming the baseline GIoU-based model. This performance gain can be attributed to the two key components of ER-IoU. First, the edge-alignment term R_{edge} directly minimizes the positional offsets of corresponding box edges, enabling more precise boundary localization without inducing unnecessary box expansion. Second, the aspect-ratio regularization term R_{ratio}, together with the adaptive weighting factor γ, further guides the network to maintain appropriate box proportions,

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 11:14:37 UTC from IEEE Xplore. Restrictions apply.

TABLE IV
EXPERIMENTAL VERIFICATION OF THE PROPOSED MODULE ON NWPU VHR-10. THE BEST RESULTS ARE MARKED IN **BOLD**

Baseline	GSAF	ER-IoU	SACL	$AP_{50:95}$	AP_{50}	Params(M)	GFLOPs
✓				61.6	94.1	20.0	59.7
✓	✓			62.6	95.2	19.6	59.1
✓		✓		62.0	95.3	20.0	59.7
✓			✓	62.7	95.1	20.0	59.7
✓	✓	✓		63.1	95.5	19.6	59.1
✓	✓		✓	62.7	95.3	19.6	59.1
✓		✓	✓	63.0	95.7	20.0	59.7
✓	✓	✓	✓	**63.4**	**96.3**	19.6	59.1

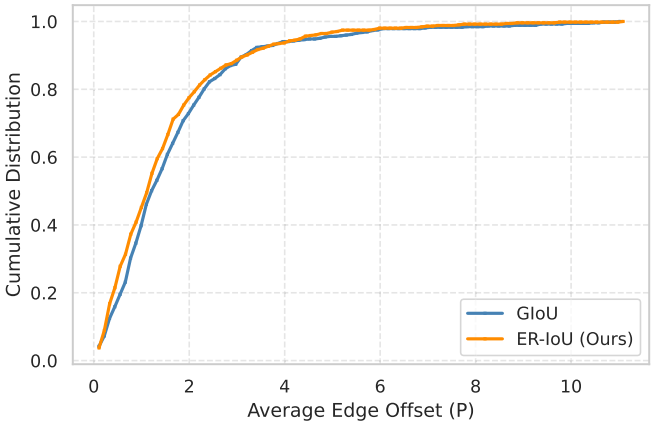


Fig. 12. CDF of $L_{\text{ER-IoU}}$ and L_{GIoU}.

especially in medium-to-high IoU regimes where fine-grained adjustments are crucial. These mechanisms jointly contribute to superior localization accuracy, as reflected by the improved AP.Besides, a detailed boundary offset analysis is conducted to assess the localization refinement capability of the proposed ER-IoU loss function. For each matched prediction and ground-truth box pair, the average edge offset P is computed as follows:

$$P = \frac{dw_l + dw_r + dh_t + dh_b}{4} \tag{22}$$

where dw_l, dw_r, dh_t, and dh_b denote the absolute deviations on the left, right, top, and bottom box boundaries, respectively. These calculated values of P are used to create a histogram and then normalized into a probability density function. The cumulative distribution function (cdf) is obtained by calculating the cumulative sum of the pdf, which is plotted in Fig. 12, comparing GIoU and ER-IoU. The y-axis represents the cumulative percentage of predictions that have an error less than or equal to the x-axis P. The results reveal that the ER-IoU loss consistently outperforms GIoU in terms of producing smaller boundary offsets across the majority of predictions. This indicates that ER-IoU effectively guides the model to produce fine-grained bounding boxes, further supporting its effectiveness in localization tasks.

To validate the convergence efficiency of ER-IoU, regression simulations are designed following a methodology similar to CIoU [42]. Target boxes with an area of 1/32 and seven

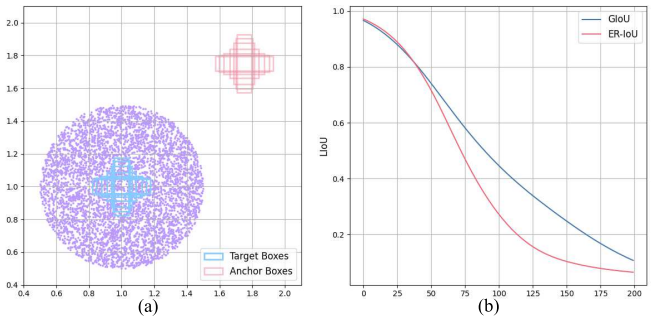


Fig. 13. Simulation experiments. (a) 1 715 000 regression cases are adopted by considering different distances, scales, and aspect ratios. (b) LIoU of different loss functions at epochs.

different aspect ratios (1:4, 1:3, 1:2, 1:1, 2:1, 3:1, and 4:1) are generated, all centered at the coordinate (1, 1). Then 5,000 anchor points are uniformly sampled within a circular region of radius 0.5 around (1, 1). For each anchor point, 49 anchor boxes are constructed by varying seven scales (1/32, 1/24, 3/64, 1/16, 1/12, 3/32, and 1/8) and seven aspect ratios (matching those of the target boxes). This setup resulted in a total of 1 715 000 regression cases (7 target boxes × 7 scales × 7 aspect ratios × 5000 anchor points), as illustrated in Fig. 13(a).

The training configuration employed the Adam optimizer with a learning rate of 0.004 and ran for 200 epochs, using LIoU as the evaluation metric. As demonstrated in Fig. 13(b), the proposed ER-IoU exhibits significantly faster convergence compared to existing methods. Meanwhile, for the ablation experiments on NWPU VHR-10 shown in Table IV, the baseline model converges at around 185 epochs, while the proposed method ER-IoU achieves convergence at approximately 165 epochs, which can also validate its superior optimization efficiency.

3) Evaluation of SACL: As shown in Table IV, the incorporation of SACL leads to an increase in AP from 94.6% to 95.4%. This improvement results from the ability of SACL to enhance ICC and interclass separability in the embedding space. In synergy with the dynamic masking mechanism based on ground-truth labels, which ensures that only queries assigned to foreground classes are involved, SACL is enabled to handle diversely distributed or weak object features and cluttered backgrounds commonly found in RSIs.

To understand how SACL influences feature representation of content queries, decoder embeddings are visualized using t-SNE, as shown in Fig. 14. After incorporating SACL, the embeddings display more distinct and compact clustering. In particular, feature points belonging to the same class demonstrate higher ICC, while different classes show better interclass separability. Notably, certain categories (e.g., AL and ship) form distinct and well-defined clusters, indicating a more structured and semantically meaningful embedding space.

Although the overall semantic separability of feature embeddings improves after integrating the SACL method, the t-SNE visualization still reveals partial overlaps between some categories. A detailed analysis of the potential causes behind

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 11:14:37 UTC from IEEE Xplore. Restrictions apply.

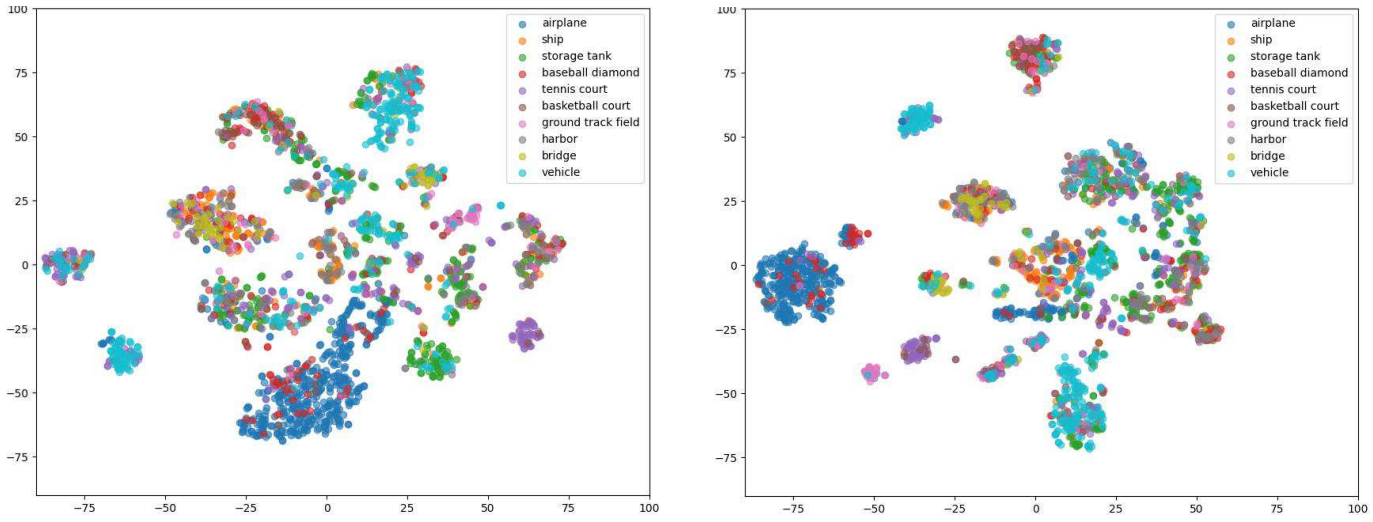


Fig. 14. Query distribution. 2D chart of the query embeddings from the first decoder layer after t-SNE dimensionality reduction. (a) Result on the baseline. (b) Result on the baseline obtained with SACL.

such feature entanglement has been conducted. First, only the backbone was pretrained on ImageNet, while both the encoder and decoder were initialized from scratch. This limits the model's ability to effectively model semantic dependencies during decoding, especially in the early decoder layers, where overlap is more likely to occur. Second, remote sensing datasets inherently contain significant interclass similarity and intraclass variation. Some categories often share similar geometric and spatial features, making them difficult to separate within a limited embedding space. In conclusion, the observed entanglement in the embedding space is a result of multiple interacting factors, including the model architecture and pretraining regime, the intrinsic properties of the dataset, and the nature of the visualization technique.

In conclusion, the feasibility and effectiveness of the proposed GSAF, ER-IoU, and SACL can be validated with the above theoretical and experimental analysis. In addition, it can be seen that the model's parameter count and computational load are reduced after applying the three methods, which further substantiate the validity of the method.

V. Conclusion

Huge scale variation and limited semantics in RSIs lead to weak feature response and increased detection complexity. To meet these challenges, S^2Net is developed for RSOD. First, the GSAF module is proposed to cope with unbalanced multiscale feature fusion. GSAF utilizes the GSA block to extract features from both big and small scales with a dual-branch structure consisting of kernels of different receptive fields, and the subsequent two PConv modules enhance local structural awareness, thereby equipping the model with the capabilities to handle multiscale objects. Second, the ER-IoU loss function is designed for the advancement of convergence and positioning, which guides anchor boxes to regress along efficient paths and fine-tune the aspect ratio. Finally, the SACL is proposed to obtain specific category foreground information from coupled features, modifying category semantic priors provided by content queries. Through extensive ablation experiments and comparison experiments on three remote sensing datasets, NWPU VHR-10, DIOR, and RSOD, the effectiveness and advantages of S^2Net can be fully validated.

References

[1] X. Zhang et al., "Remote sensing object detection meets deep learning: A metareview of challenges and advances," *IEEE Geosci. Remote Sens. Mag.*, vol. 11, no. 4, pp. 8–44, Dec. 2023.

[2] S. Gui, S. Song, R. Qin, and Y. Tang, "Remote sensing object detection in the deep learning era—A review," *Remote Sens.*, vol. 16, no. 2, p. 327, Jan. 2024.

[3] X. Fan, Z. Hu, Y. Zhao, J. Chen, T. Wei, and Z. Huang, "A small-ship object detection method for satellite remote sensing data," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 17, pp. 11886–11898, 2024.

[4] D. Zhang, Y. Ma, H. Zhang, and Y. Zhang, "Marine equipment siting using machine-learning-based ocean remote sensing data: Current status and future prospects," *Sustainability*, vol. 16, no. 20, p. 8889, Oct. 2024.

[5] C. Zheng, T. Liu, A. Abd-Elrahman, V. M. Whitaker, and B. Wilkinson, "Object-detection from multi-view remote sensing images: A case study of fruit and flower detection and counting on a central Florida strawberry farm," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 123, Sep. 2023, Art. no. 103457.

[6] S. Hinz and A. Baumgartner, "Automatic extraction of urban road networks from multi-view aerial imagery," *ISPRS J. Photogramm. Remote Sens.*, vol. 58, nos. 1–2, pp. 83–98, Jun. 2003.

[7] M. Cetin, U. Halici, and Ö. Aytekin, "Building detection in satellite images by textural features and AdaBoost," in *Proc. IAPR Workshop Pattern Recognit. Remote Sens.*, Aug. 2010, pp. 1–4.

[8] K. He, X. Zhang, S. Ren, and J. Sun, "Spatial pyramid pooling in deep convolutional networks for visual recognition," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 37, no. 9, pp. 1904–1916, Sep. 2015.

[9] S. Ren, K. He, R. Girshick, and J. Sun, "Faster R-CNN: Towards real-time object detection with region proposal networks," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 28, 2015, pp. 91–99.

[10] K. He, G. Gkioxari, P. Dollár, and R. Girshick, "Mask R-CNN," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Oct. 2017, pp. 2961–2969.

[11] J. Redmon, S. Divvala, R. Girshick, and A. Farhadi, "You only look once: Unified, real-time object detection," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 779–788.

[12] A. Bochkovskiy, C.-Y. Wang, and H.-Y. Mark Liao, "YOLOv4: Optimal speed and accuracy of object detection," 2020, *arXiv:2004.10934*.

[13] M. Yaseen, "What is YOLOv9: An in-depth exploration of the internal features of the next-generation object detector," 2024, *arXiv:2409.07813*.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 11:14:37 UTC from IEEE Xplore. Restrictions apply.

[14] R. Khanam and M. Hussain, "YOLOv11: An overview of the key architectural enhancements," 2024, *arXiv:2410.17725*.
[15] W. Liu et al., "SSD: Single shot MultiBox detector," in *Proc. Eur. Conf. Comput. Vis.* Cham, Switzerland: Springer, Oct. 2016, pp. 21–37.
[16] C.-Y. Fu, W. Liu, A. Ranga, A. Tyagi, and A. C. Berg, "DSSD: Deconvolutional single shot detector," 2017, *arXiv:1701.06659*.
[17] N. Carion, F. Massa, G. Synnaeve, N. Usunier, A. Kirillov, and S. Zagoruyko, "End-to-end object detection with transformers," in *Proc. Eur. Conf. Comput. Vis.* Cham, Switzerland: Springer, 2020, pp. 213–229.
[18] X. Zhu, W. Su, L. Lu, B. Li, X. Wang, and J. Dai, "Deformable DETR: Deformable transformers for end-to-end object detection," 2020, *arXiv:2010.04159*.
[19] F. Li, H. Zhang, S. Liu, J. Guo, L. M. Ni, and L. Zhang, "DN-DETR: Accelerate DETR training by introducing query DeNoising," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 13619–13627.
[20] Y. Zhao et al., "DETRs beat YOLOs on real-time object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2024, pp. 16965–16974.
[21] G. Cheng, J. Han, P. Zhou, and L. Guo, "Multi-class geospatial object detection and geographic image classification based on collection of part detectors," *ISPRS J. Photogramm. Remote Sens.*, vol. 98, pp. 119–132, Dec. 2014.
[22] K. Li, G. Wan, G. Cheng, L. Meng, and J. Han, "Object detection in optical remote sensing images: A survey and a new benchmark," *ISPRS J. Photogramm. Remote Sens.*, vol. 159, pp. 296–307, Jan. 2020.
[23] Y. Long, Y. Gong, Z. Xiao, and Q. Liu, "Accurate object localization in remote sensing images based on convolutional neural networks," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 5, pp. 2486–2498, May 2017.
[24] J. Yang, S. Liu, J. Wu, X. Su, N. Hai, and X. Huang, "Pinwheel-shaped convolution and scale-based dynamic loss for infrared small target detection," in *Proc. AAAI Conf. Artif. Intell.*, vol. 39, 2025, pp. 9202–9210.
[25] T. Zhao, R. Feng, and L. Wang, "SCENE-YOLO: A one-stage remote sensing object detection network with scene supervision," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5401515, doi: 10.1109/TGRS.2025.3539698.
[26] Y. Liu, X. Cheng, N. Xu, L. Wang, X. Wang, and X. Zhong, "MFAE-YOLO: Multifeature attention-enhanced network for remote sensing images object detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5631214, doi: 10.1109/TGRS.2025.3583467.
[27] T. Zhang, Y. Zhuang, G. Wang, H. Chen, L. Li, and J. Li, "A unified remote sensing object detector based on Fourier contour parametric learning," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5611225, doi: 10.1109/TGRS.2025.3540085.
[28] Y. Liu, W. Guo, C. Yao, and L. Zhang, "Dual-perspective alignment learning for multimodal remote sensing object detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5404015, doi: 10.1109/TGRS.2025.3577046.
[29] Y. Wu, Z. Li, B. Zhao, Y. Song, and B. Zhang, "Transfer learning of spatial features from high-resolution RGB images for large-scale and robust hyperspectral remote sensing target detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5505732, doi: 10.1109/TGRS.2024.3355184.
[30] B. Wang, H. Sui, G. Ma, Y. Zhou, and M. Zhou, "GMODet: A real-time detector for ground-moving objects in optical remote sensing images with regional awareness and semantic–spatial progressive interaction," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5605623, doi: 10.1109/TGRS.2025.3526799.
[31] X. Zhong et al., "Adaptive deformation-learning and multiscale-integrated network for remote sensing object detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5611619, doi: 10.1109/TGRS.2025.3541441.
[32] T. Y. Lin, P. Dollàr, R. Girshick, K. He, B. Hariharan, and S. Belongie, "Feature pyramid networks for object detection," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jul. 2017, pp. 2117–2125.
[33] S. Liu, L. Qi, H. Qin, J. Shi, and J. Jia, "Path aggregation network for instance segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 8759–8768.
[34] G. Ghiasi, T.-Y. Lin, and Q. V. Le, "NAS-FPN: Learning scalable feature pyramid architecture for object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 7036–7045.
[35] M. Tan, R. Pang, and Q. V. Le, "EfficientDet: Scalable and efficient object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 10781–10790.
[36] Y. Zhang, M. Ye, G. Zhu, Y. Liu, P. Guo, and J. Yan, "FFCA-YOLO for small object detection in remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5611215, doi: 10.1109/TGRS.2024.3363057.
[37] Y. Wu, X. Guan, B. Zhao, L. Ni, and M. Huang, "Vehicle detection based on adaptive multimodal feature fusion and cross-modal vehicle index using RGB-T images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 16, pp. 8166–8177, 2023.
[38] T. Gao, Q. Niu, J. Zhang, T. Chen, S. Mei, and A. Jubair, "Global to local: A scale-aware network for remote sensing object detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5615614.
[39] H.-I. Liu, Y.-W. Tseng, K.-C. Chang, P.-J. Wang, H.-H. Shuai, and W.-H. Cheng, "A DeNoising FPN with transformer R-CNN for tiny object detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 4704415, doi: 10.1109/TGRS.2024.3396489.
[40] J. Yu, Y. Jiang, Z. Wang, Z. Cao, and T. Huang, "UnitBox: An advanced object detection network," in *Proc. 24th ACM Int. Conf. Multimedia*, Oct. 2016, pp. 516–520.
[41] H. Rezatofighi, N. Tsoi, J. Gwak, A. Sadeghian, I. Reid, and S. Savarese, "Generalized intersection over union: A metric and a loss for bounding box regression," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 658–666.
[42] Z. Zheng, P. Wang, W. Liu, J. Li, R. Ye, and D. Ren, "Distance-IoU loss: Faster and better learning for bounding box regression," in *Proc. AAAI Conf. Artif. Intell.*, Apr. 2020, vol. 34, no. 7, pp. 12993–13000.
[43] Z. Gevorgyan, "SIoU loss: More powerful learning for bounding box regression," 2022, *arXiv:2205.12740*.
[44] S. Zhang, C. Li, Z. Jia, L. Liu, Z. Zhang, and L. Wang, "Diag-IoU loss for object detection," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 33, no. 12, pp. 7671–7683, Dec. 2023.
[45] L. Wang, Y. Zhan, L. Lan, X. Lin, D. Tao, and X. Gao, "DeIoU: Toward distinguishable box prediction in densely packed object detection," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 34, no. 11, pp. 11086–11100, Nov. 2024.
[46] C. Liu, K. Wang, Q. Li, F. Zhao, K. Zhao, and H. Ma, "Powerful-IoU: More straightforward and faster bounding box regression loss with a nonmonotonic focusing mechanism," *Neural Netw.*, vol. 170, pp. 276–284, Feb. 2024.
[47] H. Hu, X. Wang, Y. Zhang, Q. Chen, and Q. Guan, "A comprehensive survey on contrastive learning," *Neurocomputing*, vol. 610, Dec. 2024, Art. no. 128645.
[48] T. Chen, S. Kornblith, M. Norouzi, and G. Hinton, "A simple framework for contrastive learning of visual representations," in *Proc. Int. Conf. Mach. Learn.*, 2020, pp. 1597–1607.
[49] K. He, H. Fan, Y. Wu, S. Xie, and R. Girshick, "Momentum contrast for unsupervised visual representation learning," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 9726–9735.
[50] J. Li, P. Zhou, C. Xiong, and S. C. H. Hoi, "Prototypical contrastive learning of unsupervised representations," 2020, *arXiv:2005.04966*.
[51] J.-B. Grill et al., "Bootstrap your own latent—A new approach to self-supervised learning," in *Proc. 34th Int. Conf. Neural Inf. Process. Syst.*, 2020, pp. 21271–21284.
[52] F. Schroff, D. Kalenichenko, and J. Philbin, "FaceNet: A unified embedding for face recognition and clustering," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2015, pp. 815–823.
[53] P. Khosla et al., "Supervised contrastive learning," in *Proc. NIPS*, 2020, pp. 18661–18673.
[54] S. Chopra, R. Hadsell, and Y. LeCun, "Learning a similarity metric discriminatively, with application to face verification," in *Proc. IEEE Comput. Soc. Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2005, pp. 539–546.
[55] T. Jang and X. Wang, "Difficulty-based sampling for debiased contrastive representation learning," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2023, pp. 24039–24048.
[56] P. Tian et al., "Sample-level improved cross-source contrastive learning for PAN and MS joint classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5403713, doi: 10.1109/TGRS.2025.3574030.
[57] X. Wang, R. Girdhar, S. X. Yu, and I. Misra, "Cut and learn for unsupervised object detection and instance segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2023, pp. 3124–3134.
[58] M. Li, T. Jia, H. Lu, B. Ma, H. Wang, and D. Chen, "MMCL: Boosting deformable DETR-based detectors with multi-class min-margin contrastive learning for superior prohibited item detection," 2024, *arXiv:2406.03176*.
[59] H. Zhang et al., "DINO: DETR with improved DeNoising anchor boxes for end-to-end object detection," 2022, *arXiv:2203.03605*.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 11:14:37 UTC from IEEE Xplore. Restrictions apply.

[60] Y. Zhang, W. Li, W. Jia, M. Zhang, R. Tao, and S. Liang, "Cross-domain hyperspectral image classification based on bi-directional domain adaptation," 2025, *arXiv:2507.02268*.

[61] X. Ding, X. Zhang, N. Ma, J. Han, G. Ding, and J. Sun, "RepVGG: Making VGG-style ConvNets great again," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 13733–13742.

[62] L. Shi, L. Kuang, X. Xu, B. Pan, and Z. Shi, "CANet: Centerness-aware network for object detection in remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5603613, doi: 10.1109/TGRS.2021.3068970.

[63] J. Han, J. Ding, N. Xue, and G. Xia, "ReDet: A rotation-equivariant detector for aerial object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 2786–2795.

[64] J. Zhang, J. Lei, W. Xie, Z. Fang, Y. Li, and Q. Du, "SuperYOLO: Super resolution assisted object detection in multimodal remote sensing imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5605415.

[65] Q. Zhang, Y. Xu, J. Zhang, and D. Tao, "ViTAEv2: Vision transformer advanced by exploring inductive bias for image recognition and beyond," *Int. J. Comput. Vis.*, vol. 131, no. 5, pp. 1141–1162, May 2023.

[66] Y. Li, Q. Hou, Z. Zheng, M.-M. Cheng, J. Yang, and X. Li, "Large selective kernel network for remote sensing object detection," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2023, pp. 16794–16805.

[67] Z. Zhou and Y. Zhu, "KLDet: Detecting tiny objects in remote sensing images via Kullback–Leibler divergence," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 4703316, doi: 10.1109/TGRS.2024.3382099.

[68] Y. Dong et al., "Optical remote sensing object detection based on background separation and small object compensation strategy," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, pp. 1–11, 2024, doi: 10.1109/JSTARS.2024.3351140.

[69] Z. Li et al., "Context feature integration and balanced sampling strategy for small weak object detection in remote sensing imagery," *IEEE Geosci. Remote Sens. Lett.*, vol. 21, pp. 1–5, 2024.

[70] H. Yin, Z. Zhu, and H. Wang, "SED-DETR: A scale-enhanced deformable detection transformer for remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5624412, doi: 10.1109/TGRS.2025.3571033.

[71] Z. Li, Y. Wang, D. Xu, Y. Gao, and T. Zhao, "TBNet: A texture and boundary-aware network for small weak object detection in remote-sensing imagery," *Pattern Recognit.*, vol. 158, Feb. 2025, Art. no. 110976.

[72] T.-Y. Lin, P. Goyal, R. Girshick, K. He, and P. Dollár, "Focal loss for dense object detection," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Oct. 2017, pp. 2980–2988.

[73] Y. Liu, Q. Li, Y. Yuan, Q. Du, and Q. Wang, "ABNet: Adaptive balanced network for multiscale object detection in remote sensing imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5614914, doi: 10.1109/TGRS.2021.3133956.

[74] J. Zhang, A. Ding, G. Li, L. Zhang, and D. Zeng, "A pyramid attention network with edge information injection for remote-sensing object detection," *IEEE Geosci. Remote Sens. Lett.*, vol. 20, pp. 1–5, 2023.

[75] C. Zhang, K.-M. Lam, and Q. Wang, "CoF-Net: A progressive coarse-to-fine framework for object detection in remote-sensing imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5600617.

[76] Y. Zhang, T. Liu, P. Yu, S. Wang, and R. Tao, "SFSANet: Multiscale object detection in remote sensing image based on semantic fusion and scale adaptability," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 4406410, doi: 10.1109/TGRS.2024.3387572.

[77] Q. Yi, M. Zheng, M. Shi, J. Weng, and A. Luo, "AFANet: A multibackbone compatible feature fusion framework for effective remote sensing object detection," *IEEE Geosci. Remote Sens. Lett.*, vol. 21, pp. 1–5, 2024.

[78] W. Ma et al., "Adaptive feature separation network for remote sensing object detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5639717, doi: 10.1109/TGRS.2024.3454355.

[79] D. Wang et al., "MTP: Advancing remote sensing foundation model via multitask pretraining," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 17, pp. 11632–11654, 2024.

[80] Z. Li et al., "Local to global: A sparse transformer-based small object detector for remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5606516, doi: 10.1109/TGRS.2025.3525720.

[81] X. Hou, M. Liu, S. Zhang, W. Ping, B. Chen, and X. Lan, "Relation DETR: Exploring explicit position relation prior for object detection," in *Proc. Eur. Conf. Comput. Vis.* Cham, Switzerland: Springer, 2024, pp. 89–105.

[82] H. Yin, H. Wang, and Z. Zhu, "Progressive dynamic queries reformation-based DETR for remote sensing object detection," *IEEE Geosci. Remote Sens. Lett.*, vol. 22, pp. 1–5, 2025.

[83] T. Gao, Z. Li, Y. Wen, T. Chen, Q. Niu, and Z. Liu, "Attention-free global multiscale fusion network for remote sensing object detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5603214, doi: 10.1109/TGRS.2023.3346041.

[84] T. Gao, Z. Liu, J. Zhang, G. Wu, and T. Chen, "A task-balanced multiscale adaptive fusion network for object detection in remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5613515.

[85] X. Yang, S. Zhang, S. Duan, and W. Yang, "An effective and lightweight hybrid network for object detection in remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5600711, doi: 10.1109/TGRS.2023.3339624.

[86] G. Peng, Z. Yang, S. Wang, and Y. Zhou, "AMFLW-YOLO: A lightweight network for remote sensing image detection based on attention mechanism and multiscale feature fusion," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 4600916, doi: 10.1109/TGRS.2023.3327285.

[87] W. Ma, X. Wang, H. Zhu, X. Yang, X. Yi, and L. Jiao, "Significant feature elimination and sample assessment for remote sensing small objects' detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5615115, doi: 10.1109/TGRS.2024.3379436.

[88] F. Zhang et al., "Renormalized connection for scale-preferred object detection in satellite imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5638023, doi: 10.1109/TGRS.2024.3440881.

[89] T. Gao, S. Xia, M. Liu, J. Zhang, T. Chen, and Z. Li, "MSNet: Multi-scale network for object detection in remote sensing images," *Pattern Recognit.*, vol. 158, Feb. 2025, Art. no. 110983.

[90] Y. Xu et al., "From weak textures to dense arrangements: Leveraging prior knowledge for small-object detection in remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 3000915, doi: 10.1109/TGRS.2025.3564634.

[91] H. Sun, G. Yao, S. Zhu, L. Zhang, H. Xu, and J. Kong, "SOD-YOLOv10: Small object detection in remote sensing images based on YOLOv10," *IEEE Geosci. Remote Sens. Lett.*, vol. 22, pp. 1–5, 2025.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 11:14:37 UTC from IEEE Xplore. Restrictions apply.